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THE

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VOLUME THE TWENTY-THIRD.

JULY TO DECEMBER,

M.DCCC.XLIII.

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JOHN WILLIAM PARKER, WEST STRAND.

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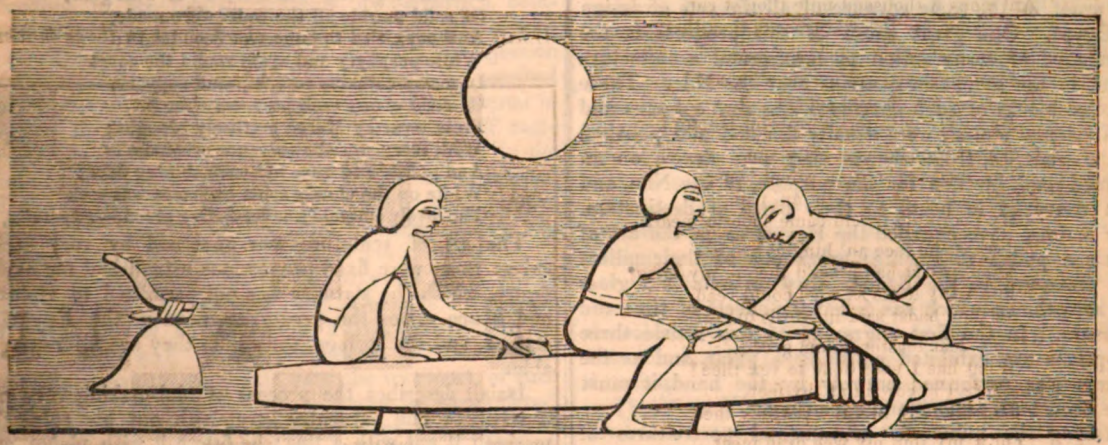
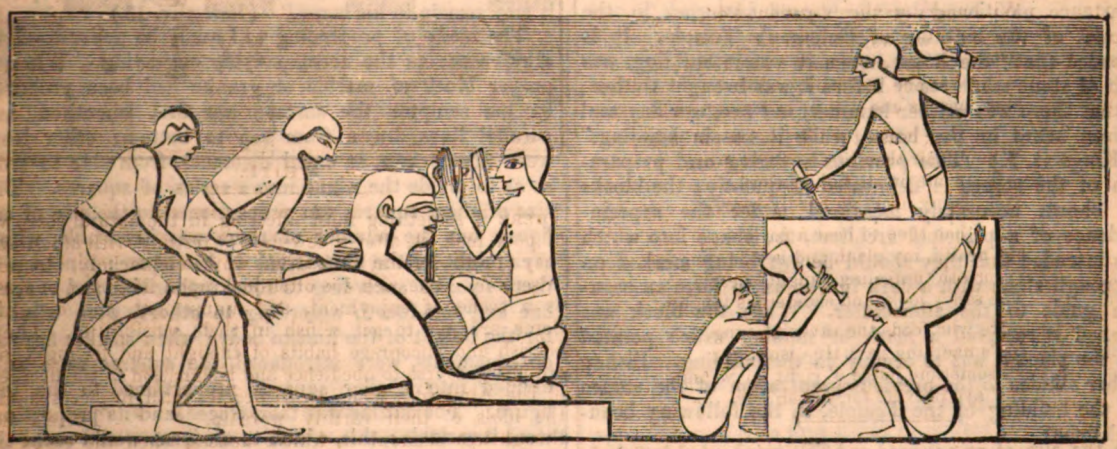
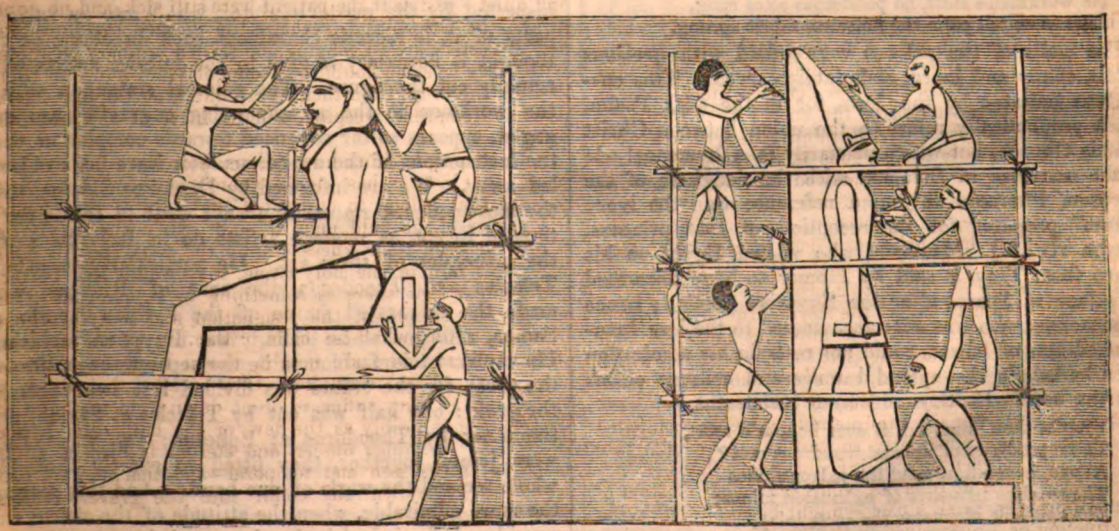
ANTIENT EGYPTIAN SCULPTURE.



ANTIENT EGYPTIANS POLISHING A PILLAR.



SCULPTURING A STATUE AND SQUARING STONES.



COLOSSAL STATUARY.

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EGYPTIAN SCULPTURE.

THE prohibition against the use of images in the second commandment prevented the Hebrews from devoting much attention to the arts of sculpture and painting; ornamental details in architecture were similarly neglected until the erection of David's palace and Solomon's Temple; indeed, the principal artists employed in both these structures appear to have been brought from Phœnicia and Egypt. The pyramids, the temples and palaces of Luxor and Karnac, "the trophied sepulchres of the kings," and the graven sarcophagi, afford full proof of the early advance of all the arts of design in Egypt. One great cause of this progress is the abundance of stone suited to ornamental architecture found in the mountain quarries near the Nile. These supplied blocks of such enormous size that columns and even colossal statues were usually finished and polished in the quarries, from whence they were drawn on an inclined road, levelled for the purpose, to the Nile, and then conveyed in boats to their place of destination. Few nations have equalled the Egyptians in the art of giving a polish to granite; specimens of their pillars now preserved in the British Museum exhibit a gloss and smoothness which has defied the effects of time and exposure. In the first engraving to this article three Egyptians are exhibited polishing a pillar, and as the process was performed entirely by the hand, it must have been both tedious and expensive.

Among the numerous monumental portraitures of the building art found on the walls of the Egyptian sepulchres, are some which explain to us a curious circumstance mentioned by the sacred historian in the account of the building of Solomon's Temple. It is recorded that "the house, when it was in building, was built of stone made ready before it was brought thither; so that there was neither hammer, nor axe, nor any tool of iron heard in the house, while it was in building." (1 Kings vi. 7.) This previous squaring and preparation of the stones is delineated frequently; the blocks are shown accurately measured under the superintendence of a principal architect, the shape into which it is intended that they should be cut being marked on the rough stone with a dark line so as to form an accurate guide to the stonecutter. When the block was finished it received a mark or number which pointed out the place it was to occupy in the building. Bishop Heber alludes to the previous preparation of the stones for the building of the Temple, in the following beautiful lines:

> . . . . . In awful state
> The Temple rear'd its everlasting gate,
> No workman's steel, no ponderous axes rung,
> Like some tall palm the noiseless fabric sprung.

The walls of the palaces were inlaid with precious metals, ebony and ivory, elaborately carved. This custom was imitated by the Jews, for in the 45th Psalm, which prophetically refers to the union between Christ and his Church, but which primarily is a hymeneal ode on the marriage of Solomon with the daughter of the Egyptian Pharaoh, we find a reference to "the ivory palaces" prepared for the reception of that princess. (Psalm xlv. 8.) Menelaus, who visited Egypt on his return from Troy, is said by Homer to have decorated his palace at Sparta after the Egyptian fashion; hence Telemachus, who was only accustomed to the less luxurious edifices of Ithaca, could not restrain his admiration when he beheld the splendid walls and ceilings of his Spartan host. He thus addresses Pisistratus,—

> View'st thou unmoved, O ever honour'd most,
> These prodigies of art and wondrous cost?
> Above, beneath, around the palace shines
> The sumless treasures of exhausted mines;
> The spoils of elephants the roofs inlay,
> And studded amber darts a golden ray.
> Such, and not nobler, in the realms above,
> My wonder dictates is the throne of Jove.—*Odyssey*, iv.

According to Lucan, the Ptolemies revived all the ancient splendour of the Pharaohs in their palatial edifices. His description of the banqueting-hall of Cleopatra gives us some very interesting information respecting the extent to which these expensive decorations were carried, and it will serve to give some notion of the gorgeous palace erected by Solomon, which was built after the model of the Egyptian Pharaohs.

> Rich as some fane by lavish zealots rear'd,
> For the proud banquet stood the hall prepared;
> Thick golden plates the latent beams infold,
> And the high roof was fretted o'er with gold;
> Of solid marble all the walls were made,
> And onyx e'en the meanest floor inlaid;
> While porphyry and agate round the court
> In massy columns rose, a proud support;
> Of solid ebony each post was wrought,
> From swarthy Meroè profusely brought;
> With ivory was the entrance crusted o'er
> And polish'd tortoise hid each shining door;
> While on the cloudy spots enchased was seen
> The lovely emerald's never-fading green.—*Pharsalia*, x.

The art of sculpture was governed by very strict rules: there were fixed proportions established for every figure, which the statuary was not permitted to violate; and hence arises the great sameness in the Egyptian statues, and the stiffness for which they are all remarkable.

Isaiah describes the process of idol-making very minutely. "The carpenter stretcheth out his rule; he marketh it out with a line; he fitteth it with planes, and he marketh it out with the compass; and maketh it after the figure of a man, according to the beauty of a man; that it may remain in his house." (Isai. xliv. 13.)

The mode of proceeding will easily be understood by a reference to the accompanying engraving. When a proper block of marble or granite had been procured by the sculptor, the surface was first smoothed, and parallel lines drawn from top to bottom; other lines were then drawn, at equal distances, from side to side, so as to divide the whole into a series of squares. The size of these squares was proportioned to the size of the figure, but the number of them was invariable, whatever might be the dimensions of the figure: nineteen of these squares, according to some authorities, and twenty-one and one-fourth according to others, were allowed for the height of the human body: when smaller figures, or ornaments, were introduced, the squares were subdivided into smaller squares, proportioned to the less figure. The outline was then traced, and its proportions were invariable: this, which to moderns would seem the most important part of the process, required no great exertion of skill in the Egyptian artist. It was then inspected by the master-sculptor, who wrote on various parts of it, in hieratic characters, such directions as he thought it necessary to give to the inferior artists who actually cut the figure. The colossal statue on which the workmen in the accompanying engraving are engaged, appears so far advanced towards completion that the instructions of the master-sculptor have been chiselled away. We are informed by Diodorus Siculus, that the most eminent statuaries always went to reside for a time in Egypt, as modern artists do in Italy, to study the principles of their art. He particularly mentions Telecles and Theodorus, the sons of Rhœcus, who made the celebrated statue of the Pythian Apollo at Samos, after, what he calls, "the Egyptian fashion." He explains this fashion to be the separate execution of the parts, for the statue was divided into two parts, at the groin: one half was cut by Telecles at Samos, and the other by Theodorus at Ephesus; yet, when they were joined together, they fitted so exactly that the whole seemed the work of one hand. And this seemed the more admirable, when the attitude of the statue was considered, for it had its hands extended, and its legs at a distance from each other, in a moving posture.

We thus see that Egyptian sculpture was almost

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wholly a mechanical process; the laws of the country prohibited the intervention of novelty in subjects considered sacred; and, the more effectually to prevent the violation of prescribed rules, it was ordained that the profession of an artist should not be exercised by any common or illiterate person. Mr. Wilkinson, indeed, has shown the great probability of the higher artists having been included in the ranks of the priesthood. In some instances, however, we find reason to believe that the Egyptian artists broke through these trammels. In the two granite statues of lions presented by Lord Prudhoe to the British Museum, we perceive a boldness and freedom of execution scarcely compatible with a strict adherence to mechanical rule. This rejection of the rigid Egyptian standard may perhaps, be ascribed to the fact of these statues having been sculptured in Upper Egypt, where the conventional rules of caste appear to have been less strictly observed than in the vicinity of Thebes and Memphis. The earlier statues of the Egyptian artists, so far as their age has been ascertained, seem to possess more freedom, though less minute finish, than those of a more recent date.

One great impediment to the progress of Egyptian art was the preference shown for colossal statues, some of which were of so stupendous a size, that even in the present day, with all the mechanical contrivances of modern ingenuity, we should view the transport of such masses as a difficulty that could scarce be surmounted. But, from the representation we have copied in the accompanying engraving of a colossal statue removed from the place in which it was wrought to that which it was destined to occupy, it appears that the Egyptians used human force only, and supplied their want of mechanical facilities for transport by employing a vast number of labourers.

We see that the colossus was mounted on a sledge and drawn along a species of railroad, previously levelled, by mere human force. A man stands on the front of the sledge pouring oil, or water, on the rails, to facilitate the transit, while another, standing on the knees of the statue, beats time with his hands, in order that all the workmen should pull together. On these occasions there was a general holiday along the road by which the statue passed; crowds of priests and warriors formed a procession in front, shouting and waving branches of palm; while relays of labourers followed behind, ready to take the places of those who might become weary. The whole representation gives a very vivid impression of the perverse idolatry of the Egyptians, and adds fresh force to the prophet's denunciation: "Thus said the Lord God; I will also destroy the idols, and I will cause their images to cease out of Noph (Memphis); and there shall be no more a prince of the land of Egypt: and I will put a fear in the land of Egypt." (Ezek. xxx. 13.)

COLOSSAL STATUE DRAWN TO ITS PLACE.

A good man submits his mind to the Governor of the whole universe, as good citizens do theirs to the law of the city.—Epictetus.

Will nothing but what God willeth, and then who can be able to hinder thee?—Epictetus.

THE FREEZING OF THE RIVER NEVA.

The River Neva, on which St. Petersburgh is situated, may be more properly considered a strait than a river. Its medium breadth is about fifteen hundred feet, while its depth in the channel is about fifty feet. It is the grand and only outlet for the waters of four great basins, each of which has an extensive natural reservoir of its own These reservoirs are the Lakes Onega, Ilmen, Saima, and Ladoga, the last mentioned being the largest lake in Europe, and receiving the superabundant waters of the other three, as well as of thirteen other streams. From such an immense accumulation of waters it is to be expected that there should be an enormous volume of water poured into the Gulf of Finland, by means of the Neva; and accordingly it has been estimated, by careful scientific measurement, that this river discharges upwards of 116,000 cubic feet of water in a second, with a velocity of about thirty-seven inches per second.

That such a body of water in rapid motion should be covered every year with a sheet of ice three feet thick, and often more, appears at first sight a very remarkable circumstance, and in order to understand it, three points have to be considered; *i. e.*, that St. Petersburgh, although not the most northern of capitals, is ascertained to be the coldest; that drift-ice is sent down in great quantities from the Ladoga; and that the winter is of very long duration.

The River Neva, at St. Petersburgh, divides itself into several branches, and across these are bridges of boats, for the purpose of communication between the two shores. Over the main trunk of the river there are also two of these bridges, one of which is on pontoons, and is managed with skill and celerity. It is nine hundred and twenty-four feet long, and is formed of fifteen pontoon boats. This bridge being cast loose at both ends, swings round against one side by the current, in about a quarter of an hour, and by means of capstans and warps is replaced in about two hours. The other cannot be managed so easily, on account of its great length, which is 2730 feet.

About the middle of November, the drift-ice is generally sent down from the Lake Ladoga. So sudden and so severe is the cold, that a cake of ice, from two to five inches thick, is formed round the edges of the northern lakes in twenty-four hours, which is almost as soon broken up by the storms to which those lakes are subject. This event is immediately announced to St. Petersburgh by telegraph. The police are then on the look-out for the appearance of the first flakes of drifted ice on the river, which is the signal for removing the bridges of boats. A resident at St. Petersburgh states, that when large sheets of ice floating down the river announce the setting in of the winter, the shorter bridges are frequently removed for a time to allow of the passage of these flakes, and then replaced, to the great inconvenience of the inhabitants of the capital. Vehicles of all descriptions are crowded together on the banks, while large boats, loaded with passengers, are seen forcing their way through shoals of drifting ice, by which they are often carried down a considerable distance. At length it becomes impossible to replace the bridge, and after a time, the passage for boats becomes more and more difficult, and is suspended altogether.

Great is the inconvenience to the capital in the interval between the removing of the bridges and the secure establishment of a road of ice across the stream. Some idea of this may be gained by supposing the river of our own metropolis to be destitute of permanent bridges, and that a similar degree of cold prevailed here. The suspension, even for a few days or a week, of the traffic which is now going on between the two shores of the Thames, would be a most serious matter; and to the inhabitants of St. Petersburgh the first freezing of the Neva is not less so. Until the ice is sufficiently thick to allow the traffic to be renewed, the inhabitants of the

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city are not only cut off from ordinary intercourse, but from all communication with the Exchange, the Custom-house, the Academies of Sciences and the Fine Arts, the Corps of Cadets of the army and navy, and other government offices, and the principal cemeteries, which are all in the Vasili Ostroff or Island, a suburb of St. Petersburgh, as Southwark is of London.

The time which is occupied before the river becomes securely frozen, depends very much on the nature of the drift ice sent down by the lake. If this has been broken into small pieces, these sometimes pass on to the gulf without encumbering the river, so that when they are all gone by, the bridge can be replaced, and things remain as before until the arrival of a fresh quantity. But if, on the contrary, the masses of ice are large, they get wedged together so firmly that they remain fixed themselves, and prevent the progress of the succeeding masses. Sometimes a violent wind will break them up, and again clear the river; but if this is not the case, the communication becomes wholly interrupted, for these masses are so fixed together as to leave large open spaces, which are very dangerous, because a thin plate of ice soon forms over them, causing them to appear like the more solid ice, while they are really insufficient to support even a foot-passenger. However, there is in a short time a sufficient thickness of ice to allow of the passage of pedestrians, who are then seen crossing the river in all directions, and in a few days more the passage is considered safe for carriages and sledges. Broad roads are then marked off by rows of fir branches placed upright, and slopes are made of planks from the quays to the ice. The river now looks like a wide and snowy valley, while the unceasing passage of carriages, sledges, merchandize, pedestrians, and troops, effectually removes the idea of the mighty stream that is rolling underneath them. This state of the river generally lasts for about five months, the general period of the breaking up of the ice being from the 5th to the 15th of April.

The great advantage which St. Petersburgh would gain in the possession of a permanent bridge, is sufficiently evident; but several circumstances unite to render it very difficult to construct one. The occurrence of the ice, the great depth of the river, and the fact that the only spot where it would be possible to build one is at a point below the dock-yards, are mentioned as the chief obstacles. A suspension bridge has been proposed by an able engineer, the design of which is very magnificent, and it is said that if executed, it would present one of the grandest monuments of the kind in the world. "The property in the Vasili Ostroff would double in value, and the inhabitants of the capital in general, while they possessed the inestimable advantage of uninterrupted communication, would enjoy the proud spectacle of their country's history, traced in the bas-reliefs of metal on the towering supports of the chains, presenting a façade of one hundred and four feet high."

THEY say of poets, they must be born such: so must mathematicians, so must great generals, and so indeed must all other denominations of men, or it is not possible they should excel. But with whatever faculties we are born, and to whatever studies our genius may direct us, studies they must still be. I am persuaded that Milton did not write his *Paradise Lost*, nor Homer his *Iliad*, nor Newton his *Principia*, without immense labour. Nature gave them a bias towards their respective pursuits, and that strong propensity, I suppose, is what we mean by genius. The rest they gave themselves.—COWPER.

POETRY reveals to us the loveliness of Nature, brings back the freshness of youthful feeling, revives the relish of simple pleasures, keeps unquenched the enthusiasm which warmed the spring-time of our being, refines youthful love, strengthens our interest in human nature, by vivid delineations of its tenderest and loftiest feelings, and through the brightness of its prophetic visions, helps faith to lay hold on the future life.

PLANT-LIKE ANIMALS.

VI.

Fig. 1.

Holothuria tubularia,
(With two of the tubercles magnified.)

IN collecting some further notices of the curious race of animals, to which we have on several occasions directed the attention of our readers, we take those which remind us of the Cactus tribe among plants. Strange and uncouth are some of the forms of cactus, and it will be owned that there is nothing very pleasing in the appearance of the animal above represented, which our fancy leads us to assimilate with them.

The *Holothuriæ* are natives of every sea, but they are particularly abundant in the Mediterranean. Some of the more remarkable species seldom come under the notice of the zoologist, on account of the great depths which they frequent, but there are others that lurk among the sea-weed, or in the crevices of the rocks, or even make holes in the sand for their habitation. They move with great reluctance, and at an extremely slow rate. In order to convey themselves from one place to another, they put forth from their mouths a bundle of tentacula or feelers, which they fix firmly on the ground, and then drag the body afterwards "at a pace slower than the shadow on the dial." By means of these feelers also, they retain themselves in one position, and keep at anchor when the sea is rough and stormy. It appears that these animals have also the power of inflating the body, and of moving their feelers so as to support themselves on the water. They swim slowly, and move their bodies in the manner of a worm.

One of the species of Holothuria is a luxury among Chinese epicures, though it is not precisely ascertained whether it is the individual above represented. Professor Jameson tells us that it is very extensively used for culinary purposes. The Chinese call it *Trepang*, and make of it rich soups, and various kinds of stews. After being cleaned and the water pressed out of it, it is laid in dry lime, called by the natives *Chanam;* afterwards it is either dried in the sun, or on stages, with wood fires under them. It forms a very considerable article of the Indian exports to China, so that there are fisheries of trepang at every country of the Indian Archipelago, from Sumatra to New Guinea. It has also been found in Ceylon and the Isle of France, and sent thence to China, and though from inexperience the inhabitants of those islands prepare it in a very inferior manner, it also finds a ready market.

The importance of this little animal in a commercial point of view, will be seen by the following facts gathered from the same authority. "The whole

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quantity of trepang sent to China from Macassar, and other parts of India, may be estimated at fourteen thousand piculs Taking this quantity at the low average of forty dollars a picul, and valuing the dollar at 4*s.* 3*d.*, its entire value in a commercial view is 119,000*l.* Notwithstanding this enormous export to China, we do not understand that its value in the market has ever been materially affected by the quantity imported; an evident proof that the demand of the market still exceeds the supply. The Celestial Empire cannot exist without its trepang and bird's nests."

There is a most extraordinary circumstance connected with the economy of the Holothuriæ, which the testimony of the most respectable observers alone enables us to believe. It is well stated by the writer of the elaborate article "Zoophytes," in the *Encyclopædia Britannica*. "When a Holothuria is placed in a basin of sea-water, it has been seen to emit jets of water from the posterior aperture at regular intervals, the jets succeeding each other, at not more than a minute's interval. This water is undoubtedly what has been rendered injurious by its stay and use in respiration, mixed probably with a considerable portion from the intestine. But when the water in the basin has become impure, these jets become also less regular; and after evidence of uneasiness, and some unusual motions, the worm will at length vomit up its tentacula, its oral apparatus, its intestine entire and with its appendages, and a large cluster, if not the whole of the ovaries. And after this complete embowelling, the animal lives for at least six or seven hours; for the empty skin shows by its motions, that nearly all its irritability remains, and even its power of locomotion is not lost. Sir John G. Dalyell, has even proved, by actual experiment, that if this poor embowelled worm is supplied with fresh sea-water at proper intervals, it will live to replace all its viscera with new growths, reproduce new tentacula, new teeth, a new stomach and intestine, and all its complicated aquiferous system, so as to be in every respect as it was previous to its wonderful evomition."

The Holothuriæ belong to Cuvier's great class *Echinodermata*, order *Pedicellata*. The form of the body is generally that of an elongated cylinder or pentagon, and the skin is sometimes scaly, like that of a fish, often of an earthy colour, or white with red spots. The shape of the body is extremely changeable in the same species. If placed in a vessel of sea-water it assumes as many varieties of size and length as a leech. This change of form in the Holothuria is said to be dependent on the action of two kinds of muscles which enter into the composition of the skin; one set forming a series of transverse parallel fibres, lining its inner surface; the other set stretching lengthwise, either in pairs or at equal distances.

Fig. 2.

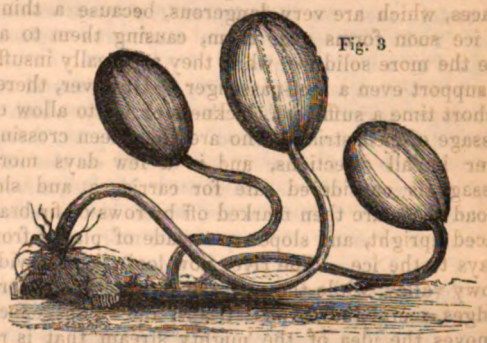
Pyrosoma elegans.

Belonging to another class is the curious species, which we may also liken to a cactus, and of which fig. 2 gives a representation, rather *smaller* than nature.

Although there is nothing in the appearance of the cylinder to convey such an idea, this mass is really composed of a great number of small animals, who unite together in this form, and swim in the ocean by their combined contraction and dilatation. Large species are often observed in the Mediterranean, arranged without much regularity. In the open sea, they assemble in considerable masses, and exhibit a sparkling phosphoric light. From the manner in which the masses are disposed in cordons, a most lively appearance is sometimes given to the sea by their phosphoric effect, resembling long trains of fire. A singular phenomenon is also attached to this phosphorescence, that is, that the colours vary instantaneously, passing from lively red to crimson, orange, green, azure-blue, and finally to an opaline yellow, when the mass appears to be in a state of repose.

These curious animals belong to the class *Acalepha*, order *Nuda*, as do also the following, which are attached to rocks and other bodies, and are deprived of all power of locomotion.

Fig. 3



Ascidia globifera.

Several of the species (of which this is one) are remarkable for the long stalks on which they are supported. They exhibit signs of life chiefly by absorbing and evacuating water. When alarmed they eject it to a considerable distance.

In considering the state of our knowledge as to the department of natural history to which these and similar animals belong, there is sufficient to humble the most diligent inquirer. The varieties among these minute inhabitants of the seas seems absolutely boundless, and it appears as if there had been scarcely a step taken by the united efforts of human industry and reason. Dr. Macculloch found in two or three days twenty undescribed species of zoophytes, and doubtless every one who would take the pains to search, would also find that much as we are indebted to zoologists for the labour with which they have classed and arranged these animals, yet that the number of unknown species is still incalculable. The view of our own ignorance should lead us often to contemplate the Divine Omniscience; but this is "to range, in imagination, the endless, boundless universe; to examine every sphere of incalculable millions; with all the mechanical actions and mathematical laws under which order is preserved; to consider every atom in each of these, and every action which is for ever taking place among them. It is, again, to bring before the mind the entire mass of the forms of organic being by which those are inhabited, and every individual in that mass, with every detail of the structure and proceedings of each, and every portion of the moral governments appointed for them, to the very individualities of their separate mental powers. The imagination cannot do this for our own globe, nor for the smallest portion of our own earth: and though it should have done it for the whole universe, it has but commenced."

It cannot be reckoned unfair to desire men, freely to give to others the liberty which they allow to themselves, if they would prove that their love of liberty is genuine and sincere.—Milner.

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ON DIORAMIC PAINTING.

I.

THE advantage of a picture over a written description is to be found, says Addison, "in the real resemblance which a picture bears to its original, which letters and syllables are wholly void of. Colours speak all languages, but words are understood only by such a people or nation."

There are but few *pictorial descriptions* more beautiful than those presented by the Diorama. In these remarkable paintings we more than realise the scenes which they are intended to present: we read the poetical language which always accompanies the successful results of the painter's art; we enjoy the harmonies of nature in the alternate effects of night and day; sunshine and shade gradually stealing over them; we are impressed with the solemnity of the Cathedral interior as the shades of evening fall slowly upon it, and it becomes resigned to solitude and gloom; objects again become visible as a few scattered lamps cast their feeble rays, and figures appear to occupy the vacant seats. These effects produced by the decomposition of form are highly successful; such, for instance, as in a recent exhibition, where rocks falling from the mountains replace the prospect of a smiling valley.

It will doubtless be interesting to the general reader to become acquainted with the means by which such beautiful effects are produced. We propose, therefore, in this article to furnish a few details on the subject, from the work of the celebrated M. Daguerre, the inventor of this branch of art, which has acquired for him almost as much celebrity as the more useful invention of the Daguerreotype. In a second article we will state the successful results of Mr George Tait's attempts to construct a *portable* diorama, whereby these peculiar effects can be produced on a smaller scale, so as to be at the command of the amateur in painting.

The diorama is a painting fitted up so as to receive light both in front and behind; so that by the full or partial admission, or the total exclusion, of one of these sources of light, a great variety of effects may be produced. No light is admitted to the eye except that which proceeds from the painting.

In Dioramic Painting the canvass is painted on both sides. Whether the subjects be illuminated by reflected or transmitted light the canvass must be very transparent, and the texture as smooth and equable as possible. Lawn or calico may be employed, but it must be manufactured of very great width, to avoid seams, which, when existing, can scarcely be concealed in the principal lights of a picture.

The canvass being properly stretched, it is primed on both sides with at least two coats of parchment size.

The first effect, which ought to be the clearer of the two, is executed on the right side of the canvass. The sketch is first made in black lead, taking care not to soil the canvass, the whiteness of which is the sole resource possessed by the artist for bringing out the lights of the picture; for white cannot be used in executing the first effect. The colours used by M. Daguerre are ground in oil, but laid upon the canvass with turpentine, to which a little animal oil is sometimes added, but only for deep shadows, and these may be varnished without injury. The method is exactly the same as in water-colour painting, with this difference only, that the colours are prepared with oil instead of gum, and applied with turpentine instead of water. The artist cannot employ white nor any opaque colour by coats, which in the second effect would occasion spots more or less tinted, according to the greater or less degree of opacity. He must endeavour to produce his effects at once—at a single stroke; because, by going over his work a second time, he injures the transparency of the canvass.

The second effect is painted on the wrong side of the canvass. In executing this, the artist must employ no other light than that which comes through the canvass, from the front part of the picture. The transparent forms of the first effect are thus seen; and these must either be preserved, or painted over according to the effect intended.

First, a wash of some transparent blue is put over the whole canvass. This coating, like the other colours, is prepared in oil, and laid on in turpentine. The marks of the brush are effaced by a huge tool of badger's skin. By means of this coating, the seams also to a certain extent are concealed by taking care to spread it thin along the selvages, which are always less transparent than the rest of the canvass. When this coating is dry, the alterations intended to be made in the first effect are sketched out.

In executing this second effect, the artist has nothing to do beyond modelling in light and shadow, without reference to local colour, or to the colours of the first picture, which are seen by transmitted light, as transparencies. This part is executed by means of a tint, of which white is the base, with which lamp-black is mixed in order to obtain a grey, the strength of which is ascertained by applying it to the wash of blue on the wrong side, and then viewing it from the right side of the picture, from which position it will not be at all perceptible if of the proper strength. The gradation of tones is produced by the greater or less opacity in this tint. It may happen that the shadows of the first effect interfere with the execution of the second. To remedy this inconvenience, and to conceal these shadows, their force can be harmonized by using the grey of a corresponding opacity, according to the strength of the shadows which it is the intention to destroy. It is necessary to urge this second effect to its utmost power.

When this general effect of light and shadow is finished on these principles, and the desired effect obtained, the picture may be coloured, the artist using only the most transparent tints prepared in oil. It is still a water-colour that is to be executed; but less turpentine must be used in these glazings, which produce a powerful effect only in proportion as they are repeated several times, and with more of oil than essence. But for slight effects of colour turpentine is sufficient.

The lighting up of the pictures. The first effect painted on the right or front of the canvass is lighted by reflection, that is to say, only by a light which comes from the front, while the second effect, or that painted on the wrong side, receives its light by transmission; that is, from behind only. In both effects both lights may be employed at once, in order to modify certain portions of the picture.

The light which gives effect to the painting in front should come from above. The illumination which falls upon the second effect—that painted behind, should proceed from vertical openings; it being always understood that these are to be completely closed when the first effect only is to be seen.

If it happen to be necessary to modify a portion in the first effect, or picture, by a light belonging to the second, that is, coming from behind, then this light must be inclosed so as not to fall except on the proper place. The windows or openings ought to be distant from the paintings at least seven or eight feet, in order to give a power of modifying the light by transmitting it through coloured media, as the exigencies of desired effects may demand. The same means are requisite for the first effect, or front picture.

To explain the principles upon which Dioramic paintings are executed and lighted up, M. Daguerre adduces the following experiment:—put upon a canvass two colours—the brightest possible—the one red, the other green, both, as near as may be, of the same intensity. Now, interpose a red medium, as a coloured glass, in the stream of light which falls upon them—what happens? The red colour reflects the rays which belong to it; the green reflecting nothing, (for it can only reflect green

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rays, and none are present,) remains black. Reverse the experiment by interposing a green glass—the effect also is reversed; the green colour gives forth its proper reflection; the red is now black. The effects, indeed, are not perfect unless the interposed media completely exclude all rays but their own, a condition not easily obtained; for coloured media have rarely the power of excluding all but one ray. The general effect, however, is sufficiently determined.

To apply this principle to dioramic paintings, though in these paintings there are only two effects represented, one of day in front, one of night behind. These effects not passing the one into the other without a complicated combination of the media which the light had to traverse, produce an infinity of other effects similar to those which nature presents in her transitions from morning to night, and the reverse. It must not be imagined that it is necessary to employ media of very intense hues in order to obtain striking modifications of colour, for often a slight shade in the medium suffices to produce a very great change in the effect.

It will thus be seen how important it is to observe the aspect of the sky, in order to appreciate the tone of a picture, whose colouring matters are subject to great decompositions. The best light for this purpose, is that from a pale sky; for where the sky is blue, it is the blue tone of the picture also, and consequently its cold tone comes out most powerfully, while its warm hues remain inactive. Their media are not present, and they are cast comparatively back into neutral tints by the blue medium of the sky, so favourable to the cold tones of the picture. It happens on the contrary when the sky is coloured that the warm tones of the picture—its reds and yellows—come forth too vigorously, and, overpowering its colder tones, injure its harmony, or it may be, give it quite a different character—a warm instead of a cold tone of colour.

These remarks go to prove that the uniform intensity of colours cannot be maintained from morning to evening—a picture cannot be the same at all hours of the day. This, perhaps, is one of the causes which contribute to render good painting so difficult to execute, and so difficult to appreciate. Painters, led into error by the changes which take place between morning and evening in the appearance of their pictures, falsely attribute these alterations to a variation in their manner of seeing, and colour falsely, while, in reality, the change is in the medium—in the light.

THERE is danger in the indulgence of feelings, let them be even the highest and the holiest of our nature, without corresponding practice to prevent their degeneration into mere aimless impulses; and these aimless impulses are found but a weak protection against the temptations that assail us in this world.

IT is true, indeed, that the Deity is more incomprehensible to us than anything else whatsoever, which proceeds from the fulness of his being and perfection, and from the transcendency of his brightness; but for the very same reason may it be said also, in some sense, that He is more knowable and conceivable than anything; as the sun, though by reason of its excessive splendour it dazzle our weak sight, yet is, notwithstanding, far more visible also, than any of the *nebulosæ stellæ*, the small misty stars. Where there is more of light, there is more of visibility; so where there is more of entity, reality, and perfection, there is more of conceptibility; such an object filling the mind more, and acting more strongly upon it. Nevertheless, because our weak and imperfect minds are lost in the vast immensity and redundancy of the Deity, and overcome with its transcendant light and dazzling brightness, therefore, hath it to us an appearance of darkness and incomprehensibility; as the unbounded expansion of light, in the clear transparent ether, hath to us the apparition of an azure obscurity; which yet is not an absolute thing in itself, but only relative to our sense, and a mere fancy in us.—CUDWORTH.

ON QUARRYING STONE.

A QUARRY is an excavation made in the ground, or among rocks, for the purpose of extracting stone for building, or for sculpture. The name appears to have originated in the circumstance that the stones, before they are removed to a distance, are first *quadrated*, or formed into rectangular blocks.

When the stone lies immediately below the surface, the first operation is to remove the earth, and lay bare the rock; but when the stone is inclosed within a hill or mountain, galleries are run into the ground, pillars of the material being left to support the superincumbent mass. Sometimes a vertical shaft is sunk, and the blocks are raised by means of a crane.

In quarrying sandstone, and those rocks which consist of regular layers, the pick, the wedge, the hammer, and the pinch, or lever, are the chief tools. But for many kinds of limestone, and for greenstone and basalt, recourse is had to the more violent and irregular effects of gunpowder. Indeed, some of the primitive rocks, such as granite, gneiss, and sienite, could scarcely be torn asunder by any other means.

The great objection to blasting by gunpowder is, that the blocks are broken irregularly, and much of the stone is wasted. Therefore, when the nature of the stone will permit, another method is adopted. A series of iron wedges are placed in a line, a few inches apart, on the natural face of the rock, and in the direction of what is called the *cleaving grain*, that is, a line parallel with the contiguous surfaces of the strata, in the direction of which the stones are more easily divided than in any other. These wedges are driven into the rock until a part becomes loosened. A channel is then cut in the direction of the length of the intended block, at a distance from the natural edge of the stone, equal to its required breadth; wedges are then driven into this channel until the stone is split in that direction also. In the hardest stones the wedges are not placed in the channels, but in *pool*, holes, as they are called, sunk in the direction in which the block is to be severed from the mass. A similar operation is then performed in the direction of the breadth of the block; and thus a large portion is detached from the original mass. The blocks are then reduced as nearly as possible to a rectangular form, by means of a tool called a *kevel*, pointed at one end and flat at the other.

A simple and efficacious method is adopted in some parts of France, for splitting large cylinders of stone into mill-stones. Horizontal indentations or grooves are chiselled out quite round the cylinder, at distances corresponding to the intended thickness of the mill-stones, and into these indentations wedges of dry wood are driven. These are then wetted, or exposed to the night dew, and next morning the different pieces of stone are found separated from each other, by the expansion of the wood, consequent on its absorption of moisture; "an irresistible natural power thus accomplishing, almost without any trouble, and at no expense, an operation which, from the peculiar hardness and texture of the stone, would otherwise be impracticable but by the most powerful machinery or the most persevering labour."

When this method of separating the blocks from the quarry is not available, on account of the hardness of the rock, gunpowder is used. This substance is simple in its application, and powerful in its effects. The grains of powder are suddenly converted into a permanently elastic air, occupying about four hundred and seventy-two times more space than their own bulk. The elastic fluid expands with a velocity calculated at the rate of about ten thousand feet per second, and its pressure or force, when thus expanding, has been estimated as equal to one thousand atmospheres, that is one thousand times greater than the atmospheric pressure upon a base of the same extent. By applying this product to a square inch, upon which the atmosphere

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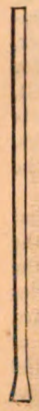
exerts a pressure of about fifteen pounds, the elastic fluid of the gunpowder will be found, at the moment of explosion, to exert a force equivalent to six tons and a half upon the square inch of surface exposed to it; and that with a velocity which the imagination can hardly follow.

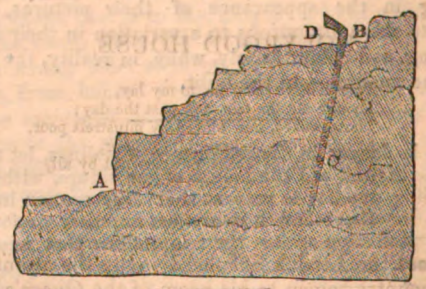
In boring a rock preparatory to blasting, it is necessary to consider the nature of the stone, and the inclination or dip of the strata, in order to decide upon the diameter, the depth, and direction of the hole for the gunpowder. The diameter of the hole may vary, according to the nature of the rock, from half an inch to two and a half inches; and the depth from a few inches to as many feet: the direction may vary to all the angles from the perpendicular to the horizontal. The tools used in this operation are very simple. The chisel, or *jumper*, as it is called, varies in size according to the work to be performed, and its edge is more or less pointed to suit the hardness of the rock to be bored. If the hole is to be small and not deep, it may be bored by a single person; with one hand he manages the chisel, which he turns at every blow so as to cross the previous cut, and with the other hand he strikes it with a hammer of six or eight pounds weight, occasionally clearing out the hole by means of a *scraper*. But when the hole is large and deep, one man in a sitting posture directs the jumper, pours water into the hole, and occasionally cleans it out, while two or three men, with hammers of ten or twelve pounds' weight, strike successive blows upon the jumper, until the rock is perforated to the desired depth. To prevent annoyance to the workmen a small rope of straw or hemp is twisted round the jumper, and made to rest in the orifice of the hole. When the holes are to be made to a greater depth than about thirty inches, it is common to use a chisel from six to eight feet in length, pointed at both ends, having a bulbous part in the middle for the convenience of holding it; it thus becomes a kind of double jumper, and is used without a hammer, with either end put into the hole at pleasure. The workmen holding this jumper by the bulbous part, lift it, and allow it to drop into the hole by its own weight, and by this simple operation a hole to the depth of five feet and upwards is perforated with ease and expedition. When the boring is completed, the fragments are carefully removed, and the hole is made as dry as possible, which is done by filling it partially with stiff clay, and then driving into it a tapering iron rod, called the *claying bar*, which nearly fills it. This being forced in with great violence, drives the clay into all the crevices of the rock, and secures the dryness of the hole. Should this plan fail, tin cartridges are used; these are furnished with a stem or tube, as shown in the following figure, through which the powder may be ignited. When the hole is dry, the charge of powder is introduced, mixed sometimes with quicklime, which, it is said, increases the force of the explosion. A long iron or copper rod, called the *pricker*, is then inserted amongst the powder, and is afterwards withdrawn, when the priming powder is introduced. The hole is filled up with burnt clay, pounded brick, stone, or any other substance not likely to produce a spark during the ramming. This is called the *tamping*. In filling up the hole, the chief danger is the production of a spark among the materials, a circumstance which has occasioned the most fatal and distressing accidents to quarriers. Prickers and rammers of copper, or of bronze, have been employed, but their greater expense, and liability to twist and break, have prevented their general introduction.



The quarrier is, of course, accustomed to suppose that the more firmly he rams in the powder the greater will be the resulting effect. It is, however, a curious property of sand, that it fills up all the void spaces in the tube or hole, and for some rocks entirely supersedes the necessity of ramming and pricking.

When the hole is fully charged with the powder and wadding, the pricker is withdrawn, and the small tubular space, or vent-hole, which it leaves, is sometimes filled up with powder; but, for the sake of economy, it is more common to insert straws filled with powder, and joined together, so as to reach the required depth. The lower straw is one terminating in the root part, where a natural obstruction occurs, or it is artificially stopped with clay to prevent the powder from being lost. The lower part of the priming straw is pared quite thin, so as to insure the inflammation of the charge of powder in the hole. Sometimes the fire is conveyed by means of the large and long green rushes, which grow in marshy ground. A slit is made in one side of the rush, along which the sharp end of a bit of stick is drawn, so as to extract the pith, when the skin of the rush closes again by its own elasticity. This tube is filled up with gunpowder; it is then dropped into the vent-hole, and made steady with a bit of clay. This being done, a slow match, called a *smift*, consisting generally of a bit of soft paper, prepared by dipping it into a solution of saltpetre, is carefully applied to the priming powder. When this match is about to be fired, the quarriers usually blow a horn or ring a bell, to give notice to all around them to retire. The explosion commonly takes place in about a minute: the priming first explodes, attended only with flame; a short interval of suspense commonly ensues; the eyes of the bystanders being anxiously directed towards the spot; the rock is instantly seen to open, when a sharp report or detonating noise takes place, and numerous fragments of stone are observed to spring into the air, and fly about in all directions, from amidst a cloud of smoke. The quarrier then returns with alacrity to the scene of his operations.

The accompanying figure shews the plan of blasting the rock, and a section of the hole ready prepared for firing. The portion of the rock to be dislodged by the explosion is that included between A. and B. The charge of powder is represented as filling the bore to C, from which point to the top, the hole is filled up with *tamping*. The *smift* is represented at D.



In the year 1831, a patent was taken out by Mr. Bickford, of Tucking Mill, Cornwall, for an invention called "the Miner's Safety Fuse." It consists essentially of a minute cylinder of gunpowder, or other suitable explosive mixture, enclosed within a hempen cord, which is first twisted in a peculiar kind of machine, then overlaid to strengthen it; afterwards it is varnished with a mixture of tar and resin to preserve the powder from moisture, and finally is coated with whitening to prevent the varnish from sticking to the fingers, or the fuses to one another. These fuses are said to have been used with good effect, and to have greatly diminished the number of accidents.

A FRENCH philosopher, whose writings are voluminous, being asked how he could find time to finish so many works, replied, "By *not* living in Paris."

JOHN W. PARKER, PUBLISHER, WEST STRAND, LONDON

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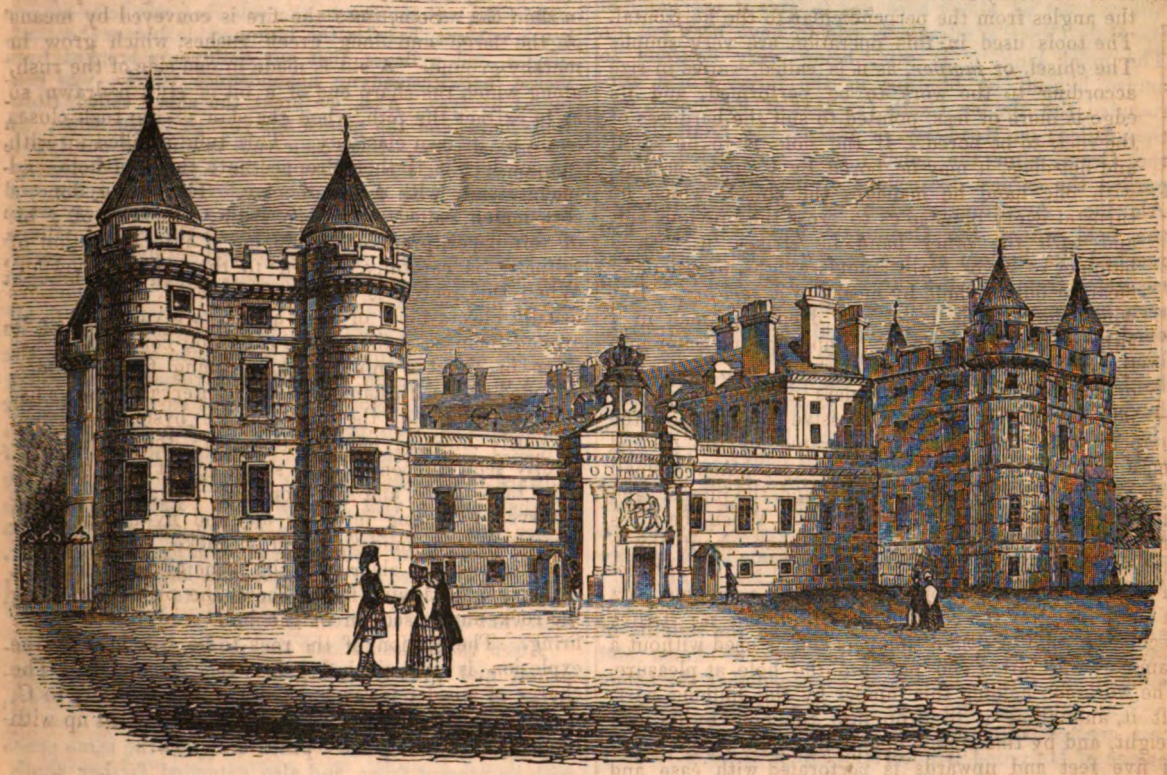
THE

Saturday Magazine.

No. 737. DECEMBER 30TH, 1843. PRICE ONE PENNY.

THE ROYAL RESIDENCES OF SCOTLAND

VI.



HOLYROOD HOUSE.

Then list, ye maidens, to my lay,
Though old the tale, and past the day;
Those wakes, now played by minstrels poor,
At midnight's darkest chillest hour,
Those humble wakes, now scorned by all,
Were first begun in courtly hall,
When royal Mary, blithe of mood,
Kept holiday at Holyrood.

THUS does the Ettrick Shepherd invite the attention of his fair countrywomen to his poem of the *Queen's Wake*, in which the different bards of Scotland meet at Holyrood House to celebrate the return of Mary to her native land. Dr. Jamieson, in his notice of this palace, truly remarks, that nothing regarding its history has given it equal interest with its being the more general residence of that beautiful queen, who, as she had been sent to France in her infancy, to avoid the rough courtship of Henry the Eighth for his son, returned from it a widow, and almost an entire stranger to that people whom she was called to govern; who, notwithstanding her natural acuteness and many accomplishments, was ill qualified for the task,—from her French education; from her early and permanent prejudices against that religion which by far the greatest part of her subjects had embraced; from the powerful influence of the Guises, the most bigoted family in Europe; from her inheriting the hereditary weakness of her name in being a prey to designing favourites; from the barbarity of the manners of even her chief nobles, whose religion had not yet taught them to be "pitiful" or "courteous;" and, perhaps, above all, from her being so unfortunate as to have a rival on the throne of England, who, unable to forgive Mary for her superiority in regard to personal charms, seems, from the hour of her return to Scotland, to have determined to subject her completely to her control. To Mary, Holyrood, from being the scene of much joy, and festivity, and folly, soon became that of deep degradation and heart-rending sorrow. Here, in her very presence, under the protection of her own apartment, her nobles entered at the instigation of her husband, and, regardless of all her entreaties, ruthlessly shed the blood of her secretary. Nothing can possibly excuse such conduct on their part; but although they had found it easy to excite jealousy in the mind of the imbecile Darnley, there is reason to believe that the actors were themselves under the influence of quite a different species of jealousy,—that of the superior intellect of the more erudite Italian.

In the second floor of the present palace are Queen Mary's apartments, in one of which her bed still remains. The embroidery on the bed and chairs is said to be chiefly the work of her own hands; and this, says Dr. Jamieson, is highly probable, as, from many specimens yet remaining, which are dispersed throughout the country, it is evident that neither she nor her maids of honour were strangers to industry. "Towards the outward door of this apartment, there are," says Arnot, "in the floor large dusky spots, said to have been occasioned by Riccio's blood staining the floor, which washing

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of the boards has not been able to take out." Pennant, after particularizing some good portraits in the other rooms, remarks that "the gallery of the palace, which takes up one side, is filled with colossal portraits of the kings of Scotland;" these are, however, for the most part imaginary.

Holyrood House, or Halyrude House, as it is called in Scotland, owes its origin to an abbey founded there in 1128 by David the First. It was denominated the Monastery of the Holy Rood, or Cross of the Craig, the latter term indicative of its situation in the vicinity of that rocky precipice now called Salisbury Craigs. To give greater celebrity to this religious foundation, as well as to influence the mind of David in determining him to erect an abbey here and to give it the name of the Holy Cross, a miracle is said to have been got up. The good king, it is said, being very desirous to amuse himself by hunting in the forest which surrounded the Maiden Castle (as that of Edinburgh was then called) on Rood-day, or that of the Exaltation of the Cross, after the solemn mass was ended, disregarded the earnest dissuasions of his devout confessor Alcwine. Accordingly when he had passed through the field where the Canongate now lies, and had reached the bottom of the crag, all his nobles being separated from him, the fairest hart that had ever been seen by human eyes, with very large antlers, so frightened the king's horse that he could not possibly restrain him. The hart followed him so hard, "that he dang baith the king and his hors to the ground." The king having thrown both his hands between the antlers of the deer, in order to save himself from its stroke, the holy cross immediately slid into his hands. The deer, of course, instantly fled and vanished from his sight "quhare now springis the Rude well." On the following night he was admonished by a vision, "to big an abbay of Channonis Regular in the same place quhare he gat the croce."

During rude times the abbeys, from the safety and accommodation which they afforded, and the sanctity of their character, became the lodgings of kings and nobles; and thus it sometimes happened that the abbey became extended into a palace, or a palatial residence was erected adjoining the abbey. The origin of the palace of Holyrood is uncertain, but it appears to have been commenced as early, at least, as the reign of James the Fourth, for his marriage with Margaret, daughter of Henry the Seventh, was celebrated in the palace of Holyrood, in August, 1503. By this marriage the Scotch prince became connected with both the White and the Red Rose of England, "an event," says Lord Hailes, "on which the fate of the two nations has turned throughout every succeeding age; to it we owe the union of the crowns, the union of the kingdoms, and the Protestant succession." Chalmers states that at the period of this marriage the palace had a chapel within it, and the chaplain was the keeper of the palace.

In the reign of Henry the Eighth the palace was burnt by the English. That monarch, in order to promote peace between the two countries, proposed the marriage of his son Edward (afterwards the sixth of that name) with the young Queen Mary. This was agreed to by the Scotch Parliament, and a treaty was entered into, but the Queen-mother, Mary of Guise, supported by Cardinal Beaton and the zealous friends of popery, determined to defeat this wise plan, and sent Mary to France. Henry was very indignant at the insult thus offered to him, and determined to avenge it in a manner such as we too often hear of in these cases, not upon the authors of the insult, but upon their unoffending subjects. The earl of Hertford was sent in 1543, with a powerful fleet, to make depredations in Scotland. The troops having been landed, marched upon Edinburgh, when a deputation from that city appeared, offering the keys of the town, on condition that the inhabitants were allowed to carry off their property and the town saved from fire. The answer was, that unless they submitted unconditionally to the will of Henry's lieutenant, "he would put them to the sword and their town to the fire." The inhabitants having displayed some feeble resistance, the following was the sad result, expressed in the words of an English writer who was on the spot.

It was determyned by the sayde Lorde-lieutenant, vtterly to ruynate and destroye the sayde towne with fyer; which, for that the nyghte drewe faste on, we omytted thoroughly to execute on that daye; but settynge fyer in thre or iiii partes of the towne, we repayred for that night vnto our campe. And the nexte mornynge very erly we began where we lefte, and continued burnynge all that daye, and the two dayes next ensuinge contynually, so that neyther within the wawles, nor in the suburbes, was lefte anyone house vnbrent, besydes the innumerable botyes [booties], spoyles, and pyllages, that our souldyours brought from thense, notwithstandyng habundaunce whiche was consumed by fyer. Also, we brent the abbey called Holy-Rode-house, and the pallice adioynynge to the same. In the meane tyme,—there came vnto vs iiii.M. of our lyghte horsemen from the borders by the Kynges Maiesties appoyntement, who dyd suche exployte in rydying and deuastying the countrie, that within vii myles euery waye of Edenborrough, they lefte neyther pyle [castle], village, nor house standynge vnbrente, nor stakes [stacks] of corne, besydes great nombres of cattayles which they brought dayley into the armey, &c.

The abbey, together with the choir and cross of its church, were destroyed in this inhuman expedition; the body of the church was all that remained standing The brazen font belonging to it was carried off by Sir Richard Lea, who presented it to the church of St. Albans, in Hertfordshire. During the minority of Edward the Sixth, after the fatal battle of Pinkie, the English paid a visit to Holyrood. Patten, who describes this visit, says, "The church and mooch parte of the house were well coouered with leade. Soon after, thei pluct [plucked] of the leade, and had down the bels, (which wear but ii) and, according to the statute did somewhat hearby disgrace the house."

After these desolations the palace was speedily repaired and greatly enlarged. The palace then consisted of five courts; viz, the west, or outermost, court, which was the largest: it was bounded on the east by the front of the palace, which occupied the same space with its present front, and also extended farther south The three remaining sides of the outer court were bounded by walls; and at the north-west corner, was a strong gate, with Gothic pillars, arches, and towers. The next court occupied the same ground with the present central court of the royal palace, and was surrounded with buildings. On the south, were two smaller courts also surrounded in the same manner; and another court on the east, bounded on the north by the chapel-royal, on the west by a line of buildings covering the same space with the present east front of the palace; on the south, by a row of buildings which are now demolished; and on the north by a wall. A writer of the year 1550, says that "the monastery of Holyrood had adjoined to it a royal palace and most pleasant gardens, enclosed by the lake at the bottom of Arthur's Seat." In the immediate vicinity of the palace there was a lion's den.

The palace was again doomed to suffer by fire. A large portion of it was burned by Cromwell's soldiers in 1650; but at the restoration it was ordered to be repaired. The present fabric was then erected. The church was also ordered to be repaired, and set apart as a chapel-royal. The manner in which this was done, has been recorded thus. A throne was erected for the Sovereign, and twelve stalls for the Knights of the Order of the Thistle; but as mass had been celebrated in it in the reign of James the Second, the populace giving vent to their fury at the Revolution, despoiled the ornaments of the inside of the church, leaving nothing but the bare walls. They even broke into the

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royal sepulchral vault; in which lay the bodies of James the Fifth; of Magdalen of France, his first queen; of the Earl of Darnley; and other of the monarchs and royal family of Scotland. James the Second, while yet only Duke of York, resided for some time at Holyrood. The palace was then privileged with a printing-press, whence a number of popish books were issued. The level strip at the bottom of the high ground behind the abbey, has received the name of the Duke's Walk, from its being a favourite promenade of James.

All that now remains of Holyrood Palace are the apartments possessed by the Duke of Hamilton, as hereditary keeper of the palace. Here the young Chevalier lodged during his residence in Edinburgh; and, a few weeks after, the Duke of Cumberland, sent in pursuit of him, occupied the very same apartment and the very same bed, which is still standing. After the defeat of the royal army at Falkirk, General Hawley quartered his troops in the gallery of the palace, and these men defaced and hewed to pieces all the representations of royalty which they found there; but the paintings have since been repaired, and are now inserted into the panels of the wainscot. These apartments also afforded an asylum to Charles the Tenth of France, then Monsieur, with a few of the emigrant nobles, between the years 1795 and 1799, when there was no safety for them in their own country. The same royal personage again found refuge for his family here, when, in 1830, he was expelled from France. In 1822, George the Fourth visited Holyrood and held his courts there, although he resided at Dalkeith, under the roof of the Duke of Buccleuch. Government has recently expended a considerable sum of money in repairing and renovating this venerable structure, and enclosing it on two sides with a magnificent iron palisade.

The precincts of the palace, including the ground which was first enclosed by James the Fifth, to the extent of about three miles, afford a sanctuary for debtors. These are, probably, the bounds of the ancient sanctuary of the monastery for the refuge and protection of criminals. This privilege is, perhaps, founded on a clause in David's charter:

I strictly forbid all persons from taking a poind [distraint], or making a seizure, in or upon the lands of the said Holy Cross, unless the Abbot refuse to do justice to the person injured.

The person who fled to the abbey was thus secure, if the Abbot chose to protect him; for no temporal judge would probably be bold enough to accuse the Abbot of injustice. Expressive of the modern indemnity, one who finds it necessary to take the benefit of the girth afforded by the environs of Holyrood House, is ludicrously styled an Abbey-laird.

The term Abbey is still popularly applied to both the chapel and the palace. Mr. Chambers describes them thus: "As they at present stand, the Palace is a handsome edifice, built in the form of a quadrangle, with a front flanked by double towers, while the abbey is reduced from its originally extensive dimensions to the mere ruin of the chapel, one corner of which adjoins to a posterior angle of the Palace. Of the palatial structure, the north-west towers alone are old." The various departments of the Palace, as well as the Chapel, are shown to strangers, for a gratuity, by the servants of the Duke of Hamilton.

In concluding these few notices of the Royal Residences of Scotland we desire to express our obligations to Dr. Jamieson's work on the Royal Palaces of that country, as also to the *Topographical Gazetteer of Scotland*, published at Glasgow during the present year; a well written work, containing abundant information on the history, topography, and statistics of most of the places of Scotland.

To describe the phenomena of the universe, to investigate their causes, and the connexion of these causes, are the principal objects of natural philosophy. To mention these objects is nearly all which is necessary to indicate its valuable effects on the mind. The habits of accurate and persevering observation, of investigation, of abstraction, and of correct reasoning, are more or less produced and cultivated by the study of this science. It furnishes abundant scope for the most sublime speculations, and calls forth the noblest exercises of the imagination, yet restrains the mind within the limits of reality. It carries us beyond the boundaries of sense, and lessens our interest in self by increasing our concern with everything around us. It enlarges the comprehension of the soul; for it offers for contemplation the laws of the universe. It prepares the student for an acquaintance with the human mind; for the strictness with which its investigations are conducted prevents that wildness of theorizing which is the bane of science; and forms the habit of cautiously attending to phenomena, in order to ascertain the general laws which regulate them. It aids the cause of religion, for it accustoms the mind to seek for the causes of observed appearances, and leads it from design and regularity to infer an intelligent First Cause.—*Encyclopædia.*

Imagine in your mind a great and powerful kingdom, or principality, in which all freely and with one consent conspire to direct their actions, agreeably to the will and command of one supreme king, the oldest and the best: and then suppose the bounds and limits of this empire not to be the river Halys, nor the Hellespont, nor the Mœotian lake, nor the shores of the ocean; but the heaven above, and the earth beneath. Here then let that great king sit immoveable, prescribing laws to all his subjects, in which consist their safety and security. And thus you see how the order and chain of this government descend down by steps and degrees, from the Supreme God to the earth and men.—Maximus Tyrius.

Other religions proposed to establish the welfare of society by positive regulations and laid down a code for the government of mankind, in all the varied walks of life; but society soon outgrew its fetters, and the code of an antiquated theocracy was thrown aside, or burst asunder by the expansion of the human mind. Christianity alone aimed at a different object. Prescribing no rule for the formation of society; dictating nothing to the forms of government, it has concentrated all its energies to coerce the human heart: it is against its depravity that all its precepts are directed; to restrain its passions that all its fetters are moulded. The consequence has been, that its progress has been as steady and progressive as that of other religions has been transient and ephemeral. Mahometanism is already falling into decay, and its gigantic frame crumbling with the corrupted mass whose energies it has confined; but Christianity, walking free and unfettered in the silver robe of innocence, adapts itself equally to all ages, and sways the heart of man alike in every period of civilization. Other religions have sought, by regulating the frame of society, to direct the human mind; but Christianity, aiming only at reforming the internal spirit of the individual, has wrought, and will for ever work, the greatest and most salutary changes on society.—Guizot.

The system of morality which Socrates made it the business of his life to teach, was raised upon the firm basis of religion. The first principles of virtuous conduct, which are common to all mankind are, according to this excellent moralist, laws of God: and the conclusive argument by which he supports this opinion is, that no man departs from these principles with impunity. "It is frequently possible," says he, "for men to screen themselves from the penalty of human laws, but no man can be unjust, or ungrateful, without suffering for his crime: hence, I conclude, that these laws must have proceeded from a more excellent legislator than man." Socrates taught, that true felicity is not to be derived from external possessions, but from wisdom, which consists in the knowledge and practice of virtue; that the cultivation of virtuous manners is necessarily attended with pleasure as well as profit; that the honest man alone is happy; and that it is absurd to attempt to separate things, which are in nature so closely united as virtue and interest.—Enfield's *History of Philosophy.*

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ON DIORAMIC PAINTING.
II.

PORTABLE DIORAMAS.

THE Diorama, being usually executed on an extensive surface of canvass, and placed in a large building fitted up for the purpose, does not seem calculated to contribute to the amusements of home, nor to exercise the talents of artists and amateurs generally. Any attempt therefore to extend the knowledge of this beautiful department of the painter's art must be received with satisfaction. Such an attempt has been made by Mr. George Tait, who has liberally communicated the details of his plan for a PORTABLE DIORAMA to the public, through the medium of the Royal Society of Arts for Scotland.

It is proper to state that Mr. Tait's first attempts to present the Diorama on a small scale were made before the publication of M. Daguerre's descriptions as detailed in a former article.

It is scarcely necessary to state that the principle of the diorama, whether on a large or a small scale, remains the same. In the one case a building is constructed expressly for the exhibition of these pictures; in the other case the pictures are contained in a small box of a simple but peculiar contrivance, furnished with openings for the admission of light before and behind, (which openings are capable of being closed by moveable shades,) and having also a small opening in front, through which the sketches may be viewed. "By this simple means" (says Mr. Tait) "many pleasing and striking effects may be produced; such as passing gleams of sunshine; day melting into moonlight; day fading into darkness, followed by morning, gradually disclosing the landscape, having its former verdure shrouded in snow."

The apparatus in detail is as follows;—

1. THE BOX. Stretching-frames are to be prepared for receiving the paper or linen on which the pictures are to be executed; and as these are confined within the narrow edges of them, the frames ought to be made thick and narrow, so as not unnecessarily to increase the width of the box, and should be bevelled off to allow access to the brush in painting the back. These frames are inserted in succession through a slit in the top of the box, about two-thirds distant from the front, and are received into a groove projecting from the top, sides, and bottom of the box, of such a breadth as fully to cover the front of them.

Two openings, one above in front, and the other behind, admit the light; and both should be as large as possible. The front opening ought to be of the form seen in the figures in order to admit the light gradually; an erect right-angle triangle H I, with its base across the breadth of the box, being placed immediately behind the front opening to aid this object. The openings have a ply of fine tissue paper, Persian silk, or other appropriate material, placed over them to diffuse the light. This is moveable, and is usually white, but may be of orange, purple, blue, or other tint, for particular purposes; and one or two plies may be used according to circumstances. The shades for the openings may be made to open and close in any manner found convenient, but so as to exclude all light when closed.

The small opening A in the front, through which the pictures are to be viewed, ought to be opposite to the height of the ordinary horizon of pictures, perhaps about a third or a fourth part of their whole height. A small tube of about two inches in length is fixed before that opening. The outer end is to be about an inch and a quarter in length, and about an inch in height, and is to be made to fit the eye, so as to screen it from extraneous light; while its inner end must expand into an oblong opening so as to allow the spectator to view the entire picture. The tube may be made to receive lenses to magnify the pictures if desired. The internal end of this tube must be so constructed as to prevent light from above shining into it.

The inside of the tube, and every part of the box seen through it, ought to be made as black as possible; for which purpose black velvet is very effectual. The rest of the inside, including the inner surface of the shades, ought to be white, in order to reflect light. The front of the box ought to be black outside, and surrounded by a black curtain, T T.

It is necessary to have the small opening and tube, shown in the figures at B, through which an exhibitor may view the pictures, when the spectator is unacquainted with the management of the apparatus. It should be on the same level as the spectator's tube. If the box be large, so as to admit of a distance of eight or nine inches from the spectator's tube, it may be on the front; if not, it will be necessary to have it on the side, close by the front, and the pictures may be reflected to it by a very small mirror within the box. This is shown at C, in fig. 3. There should be placed over and behind it, to screen the eye of the exhibitor from the light without, a black moveable shade, which may be conveniently made of two parts, the upright part to support the horizontal part when in use, and both, when not in use, to fold upon the box. The part of the box within this shade is to be painted black.

The front and back shades should be fitted up in such a manner as to be opened conveniently, either by a spectator or by an exhibitor.

The box may be made of any size or proportions—the larger, the more striking the effect. And it may be supported upon a stand, or in any other convenient manner.

2. THE PICTURES. The pictures may be either in water colour upon paper, stretched in the usual manner on the frames, or in oil upon linen.

In painting the front, whether in water colours or in oil, the lights are omitted, as in ordinary water colour painting, so as to admit the light from behind to pass through, and body colours are to be avoided. The back of the pictures is to be covered with a strong semi-transparent tint, in those parts where it is wished that light shall not pass freely, or it may be rendered opaque if required. When painting the back, no light is to be used, except that transmitted through the front.

Objects painted behind are, of course, not seen by the front light; and objects painted in front appear so faint when seen by transmitted light, that it is easy to paint the back in such a manner as to make them disappear when the back light only is admitted, by which means great changes may be produced.

The appearance of fog is produced by painting the objects intended to be affected by it, on a second surface immediately behind the front surface. Light is admitted behind. When the second surface is removed more or less from the other, the objects on it appear more or less involved in fog. And as it is brought into contact with the other, the fog appears to clear away.

A great variety of effects of daylight and moonlight may be produced by judicious management of the pictures, and by the adoption of contrivances sufficiently known, or obvious to those who have paid any attention to art generally.

3. THE LIGHT. In daylight, the back of the box is placed close to a window, and no more light ought to be admitted into the apartment than is necessary fully to light the box. At night the openings may be lighted with oil or gas, or even with a few candles if the box be small. The very strong orange tinge of ordinary artificial light is unfavourable to the natural and pleasing effect of the pictures; but it may be so far counteracted where necessary, as sometimes in night scenes or snow views, by interposing tissue paper, or other appropriate material, tinted blue.

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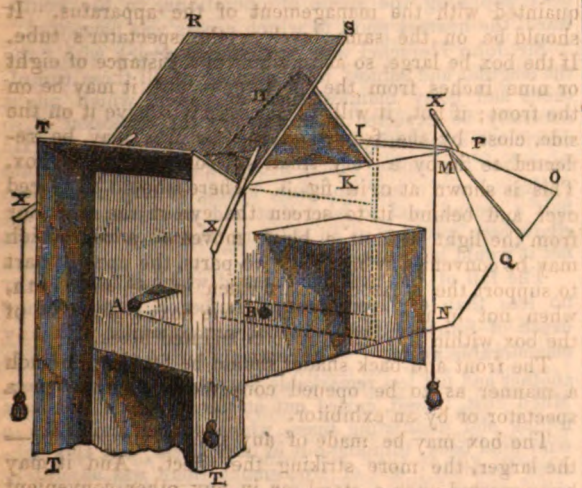
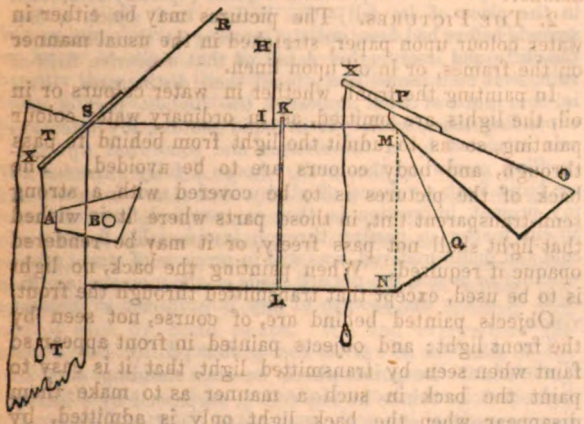
The effect produced depends in a great measure on the management of the light, and a few trials will soon enable any one to regulate its admission, so as to exhibit every change of effect.

A clearer idea of the mechanism of the box may be obtained by a summary statement of its various parts, as indicated by the letters of reference in the figures: of which No. 1 is a perspective view; No. 2, a side elevation; and No. 3, a plan of the box. The letters of reference are the same in all the figures.

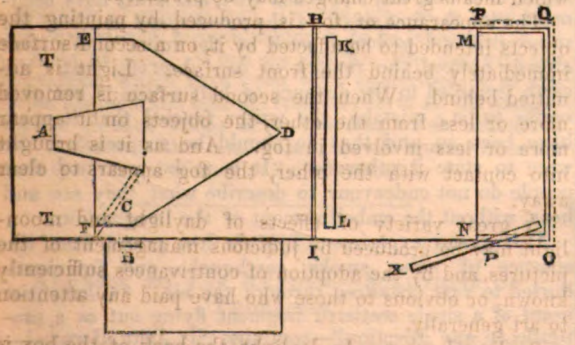
No. 1.



No. 2.



No. 3.



A. Eye-hole for the spectator.
B. Ditto for the exhibitor, with a shade over it.
C. Small mirror reflecting the picture to the exhibitor.
D E F. Form of front light.
H I. Triangle, to prevent a too sudden increase of light on raising the shade, R S.
K L. Picture in its groove.
M Q N. Back light. The slope N Q is closed; the slope M Q is open. It, and the front light, D E F, are covered with tissue paper, or other appropriate material. The back shade, P O extends beyond the opening. The intention of this construction is to admit the light in a proper position, and very gradually.
T T. Curtain hung in front, so as to shade completely the light from the spectator.

THE APPLICATION OF ELECTRICITY TO THE BLASTING OF ROCKS.

PERHAPS the greatest modern improvement that has been made in blasting rocks has been by the introduction of the galvanic battery. It is well known that by closing the circuit of a voltaic current by means of thin platinum wire, or by fine iron or steel wire, the platinum becomes red-hot, and the iron or steel becomes instantly fused. All, therefore, that is necessary is to connect the two terminal wires of a voltaic battery by means of a fine wire of platinum, or iron, and to bury this in gunpowder contained in a tin canister, or a fuse connected with a deposit of gunpowder. This was the method adopted by Colonel Pasley in removing the Royal George which lay sunk at the bottom of the water at Spithead. Canisters of gunpowder, sometimes to the extent of three thousand pounds' weight, were employed, and securely deposited in the sunken vessel, by workmen who descended in the diving bell; the terminal wires of the battery, connected as above stated, were inserted in the canisters, and these wires were then extended to a great distance from the scene of the explosion, which took place the instant they were connected with the voltaic battery. After the vessel was thus blown to pieces by repeated explosions, divers descended to clear away the wreck, and to attach guns, &c., to chains let down from a ship above, and which were then hauled up by means of a crane.

Mr. Morgan, in the *American Journal of Science*, describes a fuse or cartridge which he has used with success in connection with the voltaic battery. This cartridge is prepared by joining two pieces of clean copper wire to the ends of a fine steel wire, about one quarter of an inch in length, by means of waxed silk; a thin piece of wood is then spliced to both copper wires, to protect the steel wire from accidents, and to enable the maker to introduce it easily into a quill, or small paper tube which is to form the cartridge. This tube is filled with fine gunpowder, and made air and water-tight. Another piece of wood is then attached to this arrangement, and one of the copper wires is bent over so as to form an angle with the straight wire.

When it is required to use this cartridge the copper-wires are rubbed with sand-paper, and twisted round the wires of the voltaic battery. The cartridge is then placed deep in the hole made to receive the gunpowder, and the charge is fired from any distance.

Mr. Morgan found this arrangement very useful in removing stumps of trees; but one of his applications of it was curious and novel: he exploded some powder in a pond at the depth of ten feet, with the battery at the distance of two hundred and ten feet; the explosion, which was instantaneous, had the effect of killing a large eel, and "I have no doubt," says Mr. Morgan, "that wild fowl will yet be killed by means of shells placed at low-water on the banks where they feed; and by means of long connecting wires the shells can be made to explode simultaneously among the birds."

But the grandest application of gunpowder and the voltaic battery to the blasting of rocks was made in the month of January, 1843, at Dover. It was determined by these means to attempt the removal of an enormous mass of the cliff facing the sea, which formed an obstruction to the line of railroad. A portion of the cliff which was penetrated by the tunnel made through Shakspeare's Cliff gave way, about two years previously. About fifty yards of the tunnel were carried away, and a clear space was thus formed for the line of railroad with the exception of a projecting point, which prior to the slip alluded to was the extremity of the part of the cliff pierced by the tunnel, and to remove which was the object of the operation in question.

To clear away this mass by the tedious process of manual labour, would have cost above twelve thousand pounds, and this consideration, as well as the time that

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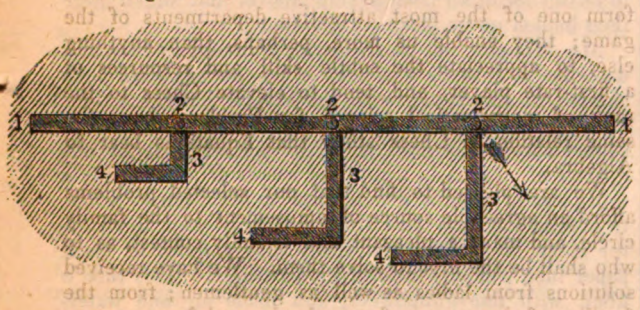
would have been lost, induced Mr. Cubitt, the engineer, to try the bold expedient of blowing it away with gunpowder. It cannot be denied, remarks Captain Stuart, whose account of this great engineering operation we follow, that there was apparent danger in the undertaking, for the weight of the mass to be removed was estimated at two million tons, and the quantity of powder used was more than eight tons, or eighteen thousand pounds. The quantity used in blowing up the fortifications of Bhurtpore was twelve thousand pounds, and this is said to have been the greatest explosion that had ever previously taken place for any single specific object.

The front of the projection was about one hundred yards wide; this front was pierced with a tunnel about six feet in height, and three in breadth; three shafts, equidistant from each other, and from the entrances to the tunnel, were sunk to the depth of seventeen feet, and galleries were run, one from each shaft, parallel with each other, and at right angles with the line of the tunnel. These galleries varied in length, the longest having been twenty-six feet, and the shortest twelve feet, and at their extremities chambers were excavated in a direction parallel with the tunnel. This description will be better understood by reference to the following figure. 1. The tunnel. 2. The shafts. 3. The galleries. 4. The chambers.



In the chambers the powder was deposited in three nearly equal quantities: it was done up in fifty pound bags, and the proportion in each chamber was contained in a wooden case, nearly as large as the chamber itself. Ignition was communicated by means of a voltaic battery; the conducting wires, one thousand feet in length, were passed over the cliff one to each chamber, and the electricity was communicated in a shed built for the purpose on the top of the cliff about fifty yards from the edge. The explosion was conducted by Lieutenant Hutchinson, R.E., who was engaged with General Pasley in blowing up the wreck of the Royal George. The time appointed for the explosion to take place, was 2 o'clock P. M., 26th January, 1843, the tide being then at its lowest ebb. The arrangements, to preserve order and prevent danger, were good. A space was kept clear by a cordon of artillery, and the following programme was issued.

"Signals, January 26, 1843.

"1st. Fifteen minutes before firing, all the signal flags will be hoisted.

"2nd. Five minutes before firing one gun will be fired, and all the flags will be hauled down.

"3rd. One minute before firing two guns will be fired, and all the flags (except that on the point which is to be blasted) will be hoisted up again."

These signals were given exactly at the specified times, and when the expected moment arrived, a deep subterranean sound was heard, a violent commotion was seen at the base of the cliff, and the whole mass slid majestically down, forming an immense *débris* at the bottom. Tremendous cheers followed the blast, and a royal salute was fired.

The remarks of different intelligent observers, as to the effects of this explosion, would of course differ according to their position with respect to the scene of explosion. One observer states that "the earth trembled to the distance of half a mile; a stifled report, not loud but deep, was heard; the *base* of the cliff, extending on either hand to upwards of five hundred feet, was shot as from a cannon, from under the superincumbent mass of chalk, seaward; and in a few seconds not less than a million tons of chalk were dislodged by the shock, and settled gently down into the sea below."

But the most eminent observer who has described the effects of this explosion is Sir John Herschel, from whose letter to the *Athenæum* we gather the following particulars. His position was on the summit of the cliff, next adjoining the scene of operations, to the southward, the nearest point to which access was permitted.

Sir John Herschel was particularly struck with "the the singular and almost total absence of all those tumultuous and noisy manifestations of power, which might naturally be expected to accompany the explosion of so enormous a quantity (19,000lbs.) of gunpowder." He describes the noise which accompanied the immediate explosion as "a low murmur, lasting hardly more than half a second, and so faint that had a companion at my elbow been speaking in an ordinary tone of voice, I doubt not it would have passed unheeded."

The fall of the cliff, the ruins of which extended over no less than eighteen acres of the beach, to an average depth of fourteen feet, was not accompanied with any considerable noise. "The entire absence of smoke was another and not less remarkable feature of the phenomenon. Much dust, indeed, curled out at the borders of the vast rolling and undulating mass, which spread itself like a semi-fluid body, thinning out in its progress; but this subsided instantly; and of true smoke there was absolutely not a vestige. Every part of the surface was immediately and clearly seen—the prostrate flagstaff (speedily re-erected in the place of its fall)—the broken turf which a few seconds before had been quietly growing at the summit of the cliff—and every other detail of that extensive field of ruin, were seen immediately in all their distinctness. Full in the midst of what appeared the highest part of the expanding mass, while yet in rapid motion, my attention was attracted by a tumultuous, and somewhat upward-swelling motion of the earth, whence I fully expected to see burst forth a volume of pitchy smoke, and from which my present impression is, that gas, purified from carbonaceous matter in passing through innumerable fissures of cold and damp material, was still in progress of escape; but whether so or not, the remark made at the moment is sufficient to prove the absence of any impediment to distinct vision."

The amount of tremor experienced by Sir John Herschel at the point where he was standing was so slight that he thinks he has felt it surpassed by a heavy waggon passing along a paved street. "The impression, slight, as it was, was single and brief, and must have originated with the first shock of the powder, and not from the subsequent and prolonged rush of the ruins." We have already noticed the remark of one observer, that "the earth trembled to the distance of half-a-mile;" but this seems to be a mistake; the writer fancied that it must have been so, and that he should be suspected if he were to state it otherwise. It is to be regretted that people do not endeavour to describe what they see and hear, without the embellishment of the imagination.

This grand experiment was no less grand from the absence of noise, smoke, earthquake, and fragments hurled to vast distances through the air. "I have not heard of a single scattered fragment flying out as a projectile, in any direction"—continues Sir John Herschel—"and altogether the whole phenomenon was totally unlike anything which, according to ordinary ideas, could have been supposed to arise from the action of gunpowder. Strange as it may seem, this contrast between the actual and the expected effects, gave to the whole scene a character rather of sublime composure than of headlong violence, of graceful ease than of struggling effect. How quietly, in short, the gigantic power employed performed its work may be gathered from the fact, that the operators themselves who discharged the batteries were not aware that they had taken effect, but thought the whole affair a failure, until re-assured by the shout which hailed its success."

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"The remarkable absence of noise and tremor which characterized this operation is explained by the structure of chalk as a material, and by the rifty state of the cliff as a body. Of all substances, perhaps chalk is the worst adapted for conveying sound, and the best for deadening the vibration propagated through it by a heavy blow. The initial hammer-like impulse of the newly-created gas on the walls of the chambers of the mines was doubtless thus deadened by traversing at least seventy-five feet of chalk, even in the shortest direction, or line of least resistance—and this must have taken place before the mass could have been sensibly moved from its seat by the expansive force generated, which however vast, proved incapable (as indeed it was expressly provided it should be,) to communicate to its enormous load any greater velocity than barely sufficient to rift and bulge it outwards, leaving gravity to do the rest. Nothing can place in a more signal light the exactness of calculation which (basing itself on a remarkably simple rule, the result of long practical experience,) could enable the eminent engineer (Mr. Cubitt,) by whom the whole arrangements are understood to have been made, so completely to task to its utmost every pound of powder employed, as to exhaust its whole effort in useful work—leaving no superfluous power to be wasted in the production of useless uproar or mischievous dispersion, and thus saving at a blow no less than 7000*l.* to the Railway Company."

O YOU unwise and unlearned! teach us first what God is that you may be believed in accusing me of impiety; tell us where God is. Is he shut up within the walls of temples? is this your piety to place God in the dark, or to make him a stony god? O you unskilful! know ye not that God is not made with hands, and hath no basis or fulcrum to stand upon, nor can be inclosed within the walls of any temple; the whole world, variegated with plants, animals, and stars, being his temple?—HERACLITUS, 500 B. C.

WHEN our object is to inquire after truth, or to teach what we know to others, we should avoid poetical comparisons, because they divert the attention, and because we are apt to fancy that we understand a subject, when we are merely pleased with the beauty or aptness of the similitude.

WE contemplate among all the diversified tribes of mankind, who are endowed with reason and speech, the same internal feelings, appetencies, aversions; the same inward convictions, the same sentiments of subjection to invisible powers, and, more or less fully developed, of accountableness, or responsibility to unseen avengers of wrong, and agents of retributive justice, from whose tribunal man cannot even by death escape. We find everywhere the same susceptibility, though not always in the same degree of forwardness, or ripeness of improvement, of admitting the cultivation of these universal endowments, of opening the eyes of the mind to the more clear and luminous views which Christianity unfolds, of becoming moulded to the institutions of religion and civilized life: in a word, the same inward and mental nature is to be recognised in all the races of men. When we compare this fact with the observations which have been fully established as to the specific instincts and separate endowments of all the distinct tribes of sentient beings in the universe, we are entitled to draw confidently the conclusion, that all human races are of one species, and of one family.—PRICHARD.

CHESS.

ADDRESS TO THE READER.—SOLUTIONS OF PROBLEMS.

BEFORE the close of the year it may be desirable to give the solutions of the Chess problems, which, from the commencement of the present volume, have been left to the ingenuity of the young student to discover. But we first avail ourselves of this opportunity of expressing our gratification at the interest which our *Easy Lessons*, and *Chess Problems*, have excited. It confirms the opinion which we stated at some length at the opening of this series*, that where Chess is introduced as an amusement into families and schools, it exerts a highly beneficial influence by exciting a taste for more exalted sources of recreation than are afforded by *games of chance*, which we regret to notice are still sometimes permitted to young people in the absence of other sedentary occupation for their leisure hours. Games of chance, so far from producing a beneficial influence on the mind, are apt to disturb the temper, excite animosity, and foster a spirit of gambling. Chess, on the contrary, is an effort of pure skill; it gives healthy exercise to the mental powers; it requires caution and forbearance on the part of both players; it leaves the victor satisfied with having won the game without the additional stimulus of "a stake;" and it entails no humiliation on the vanquished; it rather prompts him to greater exertions. We propose, therefore, in the ensuing volume of the *Saturday Magazine*, to continue our *Easy Lessons*, with the same motive that has hitherto guided us, namely, that of making the young student acquainted with a few of the leading features of the principal openings, that he may form some idea of the richness of the territory of Chess, a few of the paths to which is all that we pretend to point out to him, hoping that he may thus be induced to explore further for himself in the works of some of our best Chess writers.

It is also proposed to continue the selection of Chess Problems in the ensuing volume. Chess Problems form one of the most attractive departments of the game; they enable us more, perhaps, than anything else, to appreciate the subtle skill and resources of a first-rate player, and tend to elevate Chess to the rank of mathematical science, for Problems have the same relation to Chess study that Equations bear to Algebra.

We are pleased to find that our selected problems afford an agreeable source of amusement to the family circle, and many a pleasant and friendly contest, as to who shall be the first to solve them. We have received solutions from ladies as well as gentlemen; from the families of clergymen; from schools, and from many a solitary chess student.

In a great majority of these cases, the solutions have been quite accurate. There are however a few of our correspondents who require to be cautioned against hasty conclusions. They frequently tell us that such and such a problem cannot be solved in the prescribed number of moves,—that the problem is incorrectly printed,—that if a certain change were allowed the solution would be easy,—in short, they are anxious to escape from the conclusion that their efforts to solve the problem have failed. Our recommendation to such correspondents is to exercise a little more patience and ingenuity; and before they decide that we or the printer are incorrect, to confer with their chess-friends, and watch narrowly the locomotive powers of the Black King.

A few of our correspondents complain of difficulty in following out the moves in our *Easy Lessons*, in consequence of the concise method by which they are indicated. Our chess notation is that most commonly adopted in England; and it certainly has the merit of being simple, natural, and intelligible. But, in order to appreciate its advantages, it must be understood. A careful attention, for a few minutes, to the directions given in *Saturday Magazine*, Vol. XX., p. 93, will enable any one to become master of this notation; and he will then feel that all the artificial methods of numbering the squares, &c., tend to embarrass the student; while this notation greatly assists him in the knowledge of the structure of the board, and the relative positions of the pieces.

THE following solutions refer to Problems contained in the present volume: the page at which each Problem occurs will alone be stated.

* *Saturday Magazine*, Vol. XX., p. 92.

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PROBLEM XV., page 8.

WHITE.	BLACK.
1. Kt. to K. Kt. sixth sq. chg.	1. K. to K. Kt. square.
2. Kt. to K. seventh sq. chg.	2. K. to K. R. square.
3. R. takes K. R. P. checking.	3. K. takes R.
4. R. to K. R. square:—mate.	

If at the first move, Black take the Kt. you can then give mate in three moves: for example,

	1. K. B. P. takes Kt
2. R. takes K. R. P. checking.	2. K. takes R.
3. R. mates.	

PROBLEM XVI., page 52.

1. Kt. to Q. B. seventh sq. chg.	1. K. to Q. Kt.
2. Kt. to Q. R. sixth sq. chg. and discovering check.	2. K. to Q. R.
3. Q. to Q. Kt. eighth sq. chg.	3. R. takes Q.
4. Kt. to Q. B. seventh sq. checkmate.	

If, at the second move, Black King go to Q. B. sq., you mate at the third move with Q. at Q. B. seventh square.

It has been suggested by a correspondent that this Problem, like the well-known Problem called "Philidor's Legacy," may be solved in two moves, by playing at once Q. to Q. Kt. eighth square checking, which being taken by the Black Rook, White mates with the Knight. But suppose Black should take the Q. with his K., what then becomes of the Problem?

PROBLEM XVII., page 53.

1. Kt. takes K. B. P. checking.	1. Q. B. takes Kt.
2. Q. takes K. B. checking.	2. K. takes Q.
3. Q. B. to Q. fourth sq. chg.	3. K. to K. Kt. square.
4. Kt. to K. R. sixth sq. mate.	

If, at the first move, the Black K. go to K. Kt. sq. you play the Kt. from K. Kt. fourth sq. to K. R. sixth sq. checking: he then plays

	2. K. B. takes Kt.
3. Kt. takes B. checkmating.	

PROBLEM XVIII., page 77.

1. Q. to K. Kt. third sq. chg.	1. K. to K. fifth square.
2. Kt. to K. third sq. discovering check.	2. K. to Q. fifth square.
3. Q. takes K. P. checking.	3. Kt. takes Q.
4. B. to Q. Kt. sixth, checkmate.	

PROBLEM XIX., page 109.

1. R. takes Q. P. checking.	1. Kt. takes R.
2. Kt. from Q. Kt. eighth to Q. B. sixth square checking.	2. K. to K. square.
3. Q. to K. B. eighth sq. chg.	3. Kt. takes Q.
4. Kt. mates.	

If, at the first move, Black play his King, you mate him immediately with your Q.

PROBLEM XX., page 109.

1. Q. to Q. B. sixth sq. chg.	1. K. to Q. Kt. fifth square
2. B. to Q. R. fifth sq. chg.	2. K. takes B.
3. Q. B. P. one square.	3. Black plays.
4. Q. Kt. P. mates	

PROBLEM XXI., page 152.

1. Kt. takes K. P. checking.	1. K. B. takes Kt
2. R. to Q. B. fifth sq. checking.	2. K. B. takes R.
3. K. B. to K. fourth sq. chg.	3 Kt. takes K. B.
4. Q. P. advances, checkmate.	

PROBLEM XXII., page 224.

1. Q. to Q. sixth square.	1. K. to K. fifth square.
2. K. Kt. P. two squares.	2. K. to K. sixth square.
3. Q. B. P. one square.	3. K. to K. fifth square.
4. Q. to Q. fourth square, mate.	

SOLUTIONS TO CURIOUS CHESS PROBLEMS.

No. I., page 92.

WHITE.	BLACK.
1. Q. to K. Kt. second square.	1. R. takes Q.
2. Kt. to Q. Kt. seventh sq. mate.	

The ingenuity and difficulty of this solution are well illustrated by your first move. Its object is to prevent the Black Rook from checking your K., and also to open the square from which your Kt. gives the mate. Black has the choice of several moves: should he take your Kt. with his Kt., or your B. with his B., you checkmate with Q. at Q. second square: if he do not take the Q., but play R. to Q. seventh, you then mate with Kt. at Q. Kt. seventh as before.

No. II., page 192.

1. Q. takes P. at K. fourth sq. checking.	1. K. takes Q
2. Kt. from K. B. seventh sq. to Q. sixth square checking.	2. K. takes Kt. at Q. fourth square.
3. Q. B. P. two squares.	3. K. takes Kt. at Q. third square.
4. P. to K. eighth sq. becoming a Knight chg. and checkmating.	

The chief difficulty of this problem arises from the usual construction of the law whereby every pawn which has reached the eighth, or last, square of the Chess-board becomes promoted. The Queen being the most valuable of all the pieces, she is, of course, most generally chosen in exchange for the pawn. Indeed, this is so universally the case that it becomes engrafted, as it were, into the very language of Chess: "The Pawn moves to Queen;"—"Pawn Queens;"—"Pawn to K. B. eighth square becoming a Queen;" &c., are terms generally recognised among the technicalities of Chess, and have served to diffuse the notion that a Pawn arrived at its eighth square *must* become a Queen, provided no Queen of the same colour is on the board. But the laws of Chess recognise no such notion, and no such proviso. The Twenty-first law of the London Chess Club speaks the following unambiguous language.

Every pawn which has reached the eighth, or last, square of the Chess-board, must be immediately exchanged for a Queen, or any other piece the player may think fit, even though all the pieces remain on the board. It follows, therefore, that he may have two or more Queens, three or more Rooks, Bishops, or Knights*.

Cases do occasionally arise when the fate of a game depends on the privilege of exchanging a pawn at its eighth square for some other piece than a Queen; but as the Queen has the powers of all the other pieces except the Knight, it is, of course, unnecessary to choose a Bishop, or a Rook, (unless the player give the odds of the Queen; or undertake to mate without a Queen, &c.,) but it may be desirable to choose a Knight, on account of his peculiar checking power, and in the case before us, this very singular and beautiful problem could not be solved in four moves, but for the promotion of the pawn into a Knight instead of a Queen.

No. III., page 196

1. Kt. takes P. at K. Kt. fourth square checking.	1. B. takes Kt.
2. Kt. takes P. at K. R. third square checking.	2. B. takes Kt.
3. Q. to K. Kt. second sq. chg.	3. B. takes Q. checkmating.

If, at the second move, you play Q. to K. Kt. second square checking, the Black P. must take the Q. giving checkmate.

No. IV., page 229.

WHITE.	BLACK.
1. R. to Q. B. seventh sq. chg.	1. K. to K. R. third square.
2. R. to K. R. seventh sq. chg.	2. R. takes R.
3. Kt. to K. Kt. eighth sq. chg.	3. K. to K. Kt. fourth square.
4. K. R. P. two sq. checkmate.	

BLACK.	WHITE.
1. R. to K. R. sixth sq. chg.	1. K. takes R.
2. Q. to K. Kt. seventh sq. chg.	2. K. to K. R. fourth square.
3. Q. takes K. R. P. checking.	3. K. to K. Kt. fifth square.
4. Q. to K. R. third sq. checkmate.	

No. V., page 240.

WHITE.	BLACK.
1. Kt. to K. Kt. sixth sq. chg.	1. Kt. takes Kt.
2. Q. to K. sixth sq.	2. Black moves.
3. K. Kt. P., or the Q. checkmates.	

* The laws of the game are given in full in *Saturday Magazine*, Vol. XX., p. 247.

JOHN W. PARKER, PUBLISHER, WEST STRAND, LONDON.

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THE

Saturday Magazine.

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A BRIEF HISTORY OF THE FINE ARTS. VIII.

BELISARIUS. F. PARKER SC.

From the Engraving by Sir Robert Strange, after the Picture by Salvator Rosa.

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THE HISTORY OF COPPER-PLATE ENGRAVING.—Concluded.

Section 1.

Early British Engraving.

The art of engraving may justly be considered indigenous to British soil, since we have traces of its practice in this country from a very remote period. The rude mechanical processes of our ancestors are revealed to us by the interesting specimens discovered in Celtic and Saxon tumuli, where many of the remains of ancient weapons of war, as well as of articles evidently used for the adornment of the person, are decorated with engraving of a rude kind, executed with some tool which cannot have been very different from the graver. The numerous coins likewise discovered, are sufficient to prove the existence of that species of engraving among the ancient Britons.

The most influential patron of early British engraving was undoubtedly Alfred the Great. "Under the protection of that excellent monarch," says Strutt, "the arts began to manifest themselves in a superior degree. He not only encouraged such artists as were in England at the time, but invited others from abroad; and the works of the Anglo-Saxon goldsmiths, who were the principal engravers of that day, were held in the highest esteem upon the Continent as well as in their own country. The caskets which they made for the preservation of the relics of saints, and other purposes, were ornamented with precious stones and engravings in so excellent a style as to excite the admiration of all who saw them." That this is not an exaggerated account of early British talent, we have the surest proof in the beautiful specimen still preserved in the Museum at Oxford, and called Alfred's Jewel. This valuable work of art is of gold, adorned with enamelling and filagree work, in the midst of which is seen the half-length figure of a man, perhaps intended to represent the Saviour, or if Strutt's conjecture be right, it may rather be intended for St. Cuthbert. The back of the jewel is ornamented with engraved foliage and flowers, and its own legend informs us that it was made at the command of Alfred the Great. It was found in the Isle of Athelney, and appears to have been one of the few articles which that monarch carried thither when he retreated from his enemies.

Archbishop Dunstan is said to have worked in the precious metals, and his biographer calls him the first of engravers; but the specimen of his drawing preserved in the Bodleian Library, does not induce much regret that the works of this artist are lost to the world. The engraving of Dunstan and his contemporaries has been regarded as the degenerate issue of the art which the patronage of Alfred had called forth. The excellence of workmanship exhibited in Alfred's jewel seems to have entirely disappeared in the dark and stormy period which succeeded the reign of that monarch. Nevertheless *die-engraving*, which afforded to the Anglo-Saxon princes the means of issuing money, continued to be practised, and was thus the means of preserving and transmitting to better times the art of the engraver.

Mr. Macpherson, in his *Annals of Commerce*, enumerates various towns to which the privilege of coining was restricted by King Athelstane, and remarks that at that time (about A. D. 930) artificers would of course be found in each town capable of working in silver, and engraving the dies. Even in the more remote kingdom of Scotland there existed at that period a case for containing the Gospel, covered with silver, and having Latin verses inscribed upon it by a Scottish engraver.

The English towns to which the coinage was restricted are worthy of being named, as showing which were the places of chief importance in the kingdom. From the list we also learn that the clergy of the superior ranks shared with the king in the prerogative of coining. Cantwarabyrig (Canterbury) was to have seven coiners; namely, four for the king, two for the archbishop, and one for the abbot. Hrofreceastre (Rochester), three coiners; two for the king, and one for the bishop. Lundenbyrig (London), eight coiners. Winteceastre (Winchester), six. Lewes, two. Hæstingaceastre (Hastings), one. Cysseceastre (Chichester), one. Hamtun (Southampton), two. Werham (Wareham), two. Exanceastre (Exeter), two. Sceaftsbyrig (Shaftesbury), two. Other burgs, not named, one each. Canute increased the number of coining places to thirty-seven, adding to those above named the following, of which we give the modernized names only for the sake of brevity. Bristol, Cambridge, Chester, Colchester, Crookhorn, Derby, Dorchester, Dover, Gloucester, Hereford, Hertford, Hythe, Huntingdon, Ipswich, Leicester, Lincoln, Manchester, Norwich, Nottingham, Oxford, Sarum, Shrewsbury, Southwark, Stamford, Sudbury, Tamworth, Thetford, Wells, York.

The art of die-engraving seems to have given rise to the use of seals. Before their introduction lands were sometimes disposed of by word of mouth, sometimes by laying a turf of the land with solemnity on an altar dedicated to sacred purposes, sometimes the lord gave the tenant a sword, bow, helmet, or drinking-horn, to certify the transfer of the property. Formerly there was in the possession of a gentleman in Berkshire a drinking-horn, on which was engraven, "I King Knoute (Canute) have given thee this horn to hold thy land by."

When at length seals came into use, they appear to have been made of brass, and the impression was taken on lead instead of wax, so that a forcible blow was necessary, as in the case of coining, to produce the required impression. It appears that at the arrival of the Normans in England, they, disliking the English manner of ratifying their charters, ordered them to be confirmed by impressions on wax from the special seals of the parties concerned. The only seal in England that is known to have been impressed on wax before this period, is the seal of Edward the Confessor, who was educated in Normandy, and who probably acquired the custom in that country.

Section 2.

Engraving on Monumental Brasses.

Soon after this period a new and more simple species of engraving was either introduced or invented in England, for specimens of which we may look in the more important of our old country churches without much fear of being disappointed. The brass plates which resound beneath our feet as we tread the aisles of some ancient edifice, contain the specimens in question. The engravings on these plates are executed entirely with the graver, the shadows being expressed by lines or strokes strengthened according to the depth of shade required, and occasionally crossed with other lines a second or a third time, in the same manner that a copper-plate is engraved that is intended for printing. On some occasions a mere outline has been cut. "These engraved effigies," says a writer in Rees's *Cyclopædia*, "are often found cemented on those horizontal tomb-stones, which form part of the pavement within the churches, where the feet of the congregation, which kept the light parts bright by friction, filled the incisions with dust, and thus darkened the shades. Very neat or exquisite workmanship was not therefore aimed at, and is not to be expected; yet some of these engravings bear no small evidence of the abilities of those by whom they were produced, and considering the dark period during which they were executed, are entitled to more praise than many engravings which have been subsequently produced. The engraver's style of drawing the human figure differs little from that of the contemporary illuminators of missals, and though the hands, feet, and other nudities are rarely tolerable, the stiffness of the draperies does in many instances bear considerable resemblance to the tissued and embroidered vestments of the entombed abbots and other dignitaries of the Roman Catholic church; while the faces occasionally display attempts not altogether successless, as it should seem, at individual portraiture."

In illustration of the above remarks, a specimen of the monumental brasses is given in the same work. It is from the tomb of William de Fulbourn, in Fulbourn Church, Cambridgeshire. This individual was appointed a baron of the exchequer in 1328, and is represented in the effigies of an ecclesiastic in a richly ornamented cope under an elegant canopy. The pattern of the embroidery is not inelegant, and the ornamental designs are engraved with considerable care. Thus were the engravers of England from the time

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of the Crusades well acquainted with an art which seemed to verge so closely on that of taking impressions on paper, that we are astonished that the latter was not sooner discovered. Whatever may have been the circumstance that led to that discovery, we are well assured that having once been made it must have spread widely and rapidly; yet we have no account of its introduction into England.

Section 3.

Early Modes of Illustrating Books.

Walpole, in his *Anecdotes of Painting*, devotes a volume to English engravers. He remarks that to a certain degree engraving was employed here almost as soon as printing, for many printers used small plates for their devices and rebuses. Caxton's *Golden Legend*, printed in 1483, is ornamented with a group of saints, and many other cuts dispersed through the body of the work. Other works of Caxton and of his successor Wynkyn de Worde, are likewise decorated with cuts. The same practice was common among subsequent printers. "One considerable work," says Walpole, "published by John Rastell, was distinguished by prints of uncommon merit for that age. It was called *The Pastyme of the People*, and by Bishop Nicholson, in his 'Historical Library,' *Rastell's Chronicle*. This scarce book, of a very large size, I saw at the auction of Mr. Ames's library; it had many cuts, eighteen of which were in great folio, representing the kings of England, so well designed and boldly executed as to be attributed to Holbein, though I think they were not of his hand." Grafton's *Chronicle*, printed in 1569, contained many heads, as of William the Conqueror, Henry the Eighth, and Queen Elizabeth. Yet though portraits were used in books, there is no trace of single prints being wrought off in that age. Henry the Eighth was in possession of some prints, which are enumerated in the catalogue of his goods, &c., but these were probably the production of foreign artists. At that time it appears that a painting was called "a table with a picture;" a print was termed "a cloth stained with a picture;" and models and bas-reliefs were called "pictures of earth." Thus the inventory referred to contains such items as the following. "Item. One table, with the whole stature of my lord prince his grace, stained upon cloth, with a curtain." "Item, One cloth, stained with Phebus rideing with his cart in the air, with the history of him."

The first book that appeared with cuts from copper-plates, appears to have been the *Birth of Mankind, otherwyse called, The Woman's Book*, dedicated to Queen Catherine, and published by Thomas Raynalde, in 1540, with many small copper cuts, but to these no name was affixed. Walpole well remarks on the discovery of printing and engraving, "It is amusing to observe what obvious arts escape our touch, and how quickly various channels are deduced from a source when once opened. This was the case of the press. Printing was not discovered till about the year 1430: in thirty years more it was applied to the multiplication of drawings. Authors had scarcely seen that facility of dispersing their works, before painters received an almost equal advantage. To each was endless fame in a manner ensured, if they had merit to challenge it. With regard to prints, the new discovery associated the professors, in some degree, with the great masters whose works they copied; thus forming an intimate acquaintance between painters and engravers."

Section 4.

The first English Engraver.

The first English engraver of whom we have an account is Thomas Geminus, or Geminie. He executed the copper-plate engravings which appeared in Vesalius's *Anatomy*, printed in England, in 1545. The first edition was dedicated to Henry the Eighth, and was written in Latin. Geminus afterwards published a translation by Nicholas Udal, and dedicated it to Edward the Sixth. The preface, written by the translator, contains the following passage.—"Accepte therefore, jentill reader, this Tractise of Anatomie, thankfully interpreting the labours of Thomas Gemini, the workman. He that with his great charge, watch, and travayle hath set out these figures in pourtrature, will most willingly be amended, or better perfected of his own workmanship, if admonished." This Geminus was a printer as well as an engraver, dwelling at Blackfriars, where he published a *Prognostication*, &c., relating to the weather, the phenomena of the heavens, &c., with a number of cuts. "So congenial an art as engraving," says Walpole, "when once discovered, could not fail to spread in an age of literature. That accomplished prelate, Archbishop Parker, who thought that whatever tended to enlighten and civilize the human mind was within his province, seems to have been the most conspicuous patron of the arts in the reign of Elizabeth."

The queen, it appears, was too economical, or too deficient in taste, to give full encouragement to genius, but the riches and flourishing condition of the country offered sufficient invitations to the arts. Archbishop Parker employed in his palace at Lambeth a printer, a painter, and two or three engravers. Above twenty books were published by this prelate from his own printing-establishment. The chief of the engravers employed by him was Remigius Hogenbergh, who twice engraved a portrait of the archbishop. At Ruckolt, in the parish of Low-Layton, in Essex, was a large genealogy of the kings of England, from the Conquest to Queen Elizabeth, with all the lines of France and England, under these two titles, *Linea Valesiorum, et Linea Angliæ*. This genealogic chart was inscribed with the name of Hogenbergh. This engraver had a brother, Francis Hogenbergh, who executed a print of Queen Mary the First, dated 1555, in allusion to her era, but supposed to be done after her death. In 1570 and 1572 expensive works printed at Cologne and Antwerp were partly illustrated by English artists, and as Walpole observes, engraving was on no contemptible footing in England when we had professors worthy of being employed on Flemish editions; for Flanders was at that time a capital theatre of arts and learning.

Section 5.

The First Maps of Counties.

An early engraver deserving particular notice was Christopher Saxton, who enlarged the geographical knowledge of his day by drawing and engraving the first maps of our counties ever seen in England. This man raised himself from the condition of a servant to a position of considerable eminence by his peculiar talents. Those talents, were, however, fostered and brought to light by the generous patronage of his master, Thomas Sekeford, Esq., Master of Requests, and Master of the Court of Wards. This gentleman encouraged Saxton to publish a complete set of maps of the counties of England and Wales, and was himself responsible for the cost attending their publication. Many of these maps were engraved by Saxton himself; but in some of them he was assisted by Hogenbergh, Ryther, and others.

The labour of six years was bestowed upon these maps, and they appeared in 1579, dedicated to the Queen, and adorned with the royal arms, and also with those of the generous Sekeford. Camden and Thoresby commend Saxton's work, and speaking of his map of York declare it to be the best map of the county ever published. This rare map is adorned at the corners with views of the city of York and the town of Hull. Augustine Ryther assisted Saxton in the engraving of it.

There was but little approach to neatness or beauty of style in the engravings of English artists at this period, and so little variety is there in the manner of their engraving for a considerable period, that we find scarcely more than a dry catalogue of names. Reginald Elstracke, who lived at the close of the sixteenth and commencement of the seventeenth centuries, is commended for greater neatness than some of his predecessors. But a family named Passe, who came hither from Utrecht, early in the seventeenth century, have the credit of introducing a better taste and more elaborate workmanship. Crispin de Passe was a man of letters, not only industrious to perfect himself in his art, but fond of promoting it in others. In a drawing-book, published by him in 1643, he discovers some knowledge of geometry and perspective, gives directions for the proportions of the human body, for drawing in the academy by lamplight, and for the use of the lay figure in studying draperies; he details the proportions of horses, lions, and other quadrupeds, and also of birds and fishes. This artist was known and encouraged by Rubens, and other great men of the day, and his copy of the female proportions employed by Rubens is evident in his works. Strutt's opinion of this artist is as follows. "Excepting some little stiffness which occasionally appears, and the want of harmony with respect to the distribution of the light and shadow, (a fault that prevailed at the time in which he lived,) his best works possess a very considerable share of merit, especially his portraits, many of which he drew from the life, and the far greater portion of his historical and emblematical subjects

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are engraved from his own compositions. He drew the human figure correctly, and marked the extremities with a degree of exactness not usually seen in the works of those masters who employed themselves upon small subjects."

Four of the children of Crispin Passe became known as artists, among whom was Magdalen Passe, the first female engraver we read of as practising the art in England. Few of her productions remain to us; but it appears that she worked in a free style, without high finish. Walpole mentions a very scarce little head executed by her, which he had in his own possession. It represented the Lady Katherine, at that time marchioness, afterwards duchess, of Buckingham, with a feather fan. One of the pupils of this family was a man of talent, and the mantle of the Passes seemed to have descended on his shoulders; but he was of an indolent and careless disposition, and therefore the opportunities of advancement which were afforded him, did not save him from dying in indigence and neglect. This was John Payne, recommended to Charles the First, and esteemed a man of decided genius; but his works are not numerous; for he led an irregular life, and died early. He executed frontispieces and portraits for books; and also other subjects, landscapes, animals, flowers, fruit, birds, &c. Several of his portraits are very finely executed, and may be reckoned as the best of his works, being performed entirely with the graver in a free open style, with a pleasing effect. This artist engraved a very large print of a ship, called the *Royal Sovereign*, of two plates, which when united, were three feet long by two feet two inches high. He died in 1647-8, in his fortieth year.

Section 6.

Appointment of an Engraver Royal.

Charles the First was a general patron of the arts, and promoter of literature, and he is honourably mentioned as the first English monarch who was sufficiently sensible of the beauty of engraving, and of the popular and important purposes to which it might be applied, to appoint an engraver royal. "The king took care to see that the title was not worthlessly thrown away in mere kindness to a courtier, and Voerst was the first artist on whom that honour was conferred. He was a native of Holland, and in what year he came hither is uncertain. He was an able draughtsman, and hatched his drawings with pen and ink, as was the general custom of the artists, particularly the engravers of that period." This artist engraved many of Vandyke's portraits, and among the rest, the celebrated portraits of Charles the First and his queen; with those of Sir Kenelm Digby, Inigo Jones, and others. A competitor for public fame, and also for the favour of the king, was Lucas Vorsterman, whose works are considered by some writers as superior to those of Voerst. He is said to exhibit more careful finish and painter-like feeling, and to have been to Rubens and Vandyke in England, what Mark Antonio was to Raphael in Italy. Mr. Strutt considers that no one ever engraved more successfully from the pictures of Rubens, and remarks that Vorsterman, although a master of the graver, and able to handle it with the greatest facility, yet paid much greater attention to the general effect of his prints than to the regularity of his strokes, wishing apparently to enter into the thoughts of the master, and transcribe on copper the very life and spirit of his pictures, rather than to show his own dexterity in the mere mechanical part of the workmanship. This artist was a native of Antwerp, and coming over to England in 1623, he was occasionally employed both by Charles the First, and the Earl of Arundel.

The celebrated Vandyke himself contributed to advance English engraving by his vigorous and expressive etchings, which excited the surprise and emulation of his contemporaries.

Here we must also mention a meritorious, but ill-fated artist, who was one of the first who practised the art of etching in England. This was Wenceslaus Hollar, a gentleman by birth, and a native of Prague, in Bohemia. He was brought up to the profession of the law, but the miseries of civil war, which led to the celebrated battle of Prague, obliged him to leave his home at an early age, and take refuge at Frankfort, where he studied under an able designer and engraver. He soon attained considerable skill, and being afterwards introduced to the Earl of Arundel at Cologne, he accompanied that nobleman to England, and was recommended to the notice of Charles the First. He does not appear to have met with many admirers in England, and the political disturbances of this country soon destroyed the hopes he might have formed on his arrival. On account of his attachment to the royal cause, he was imprisoned at Basing House, in Hampshire, and on obtaining his liberty was obliged to repair to Antwerp. Revisiting England in 1652, he met with employment, but barely earned a subsistence. He used to work for the mercenary book and printsellers at the rate of fourpence an hour, always having an hour-glass before him; and he was so scrupulously exact, that when at meals, or even when talking with the persons for whom he was working, and about their own business, he constantly laid the glass on its side to prevent the sand from running. Exemplary industry, and rigid adherence to principle, did not rescue Hollar from the miseries attendant on poverty. The return of his friends at the Restoration somewhat bettered his circumstances for a time. He was employed by government to make drawings of the town and forts of Tangiers, and in this perilous service he narrowly escaped being killed or made prisoner by the Turks. On his return to England, after one year's absence on this arduous service, all he received was one hundred pounds, and this was only paid after long delay, and many humble solicitations. Hollar is described as a meek-tempered man, wanting in necessary confidence in his own powers, though a person of superior talent. On his death-bed, at the age of seventy, he is said to have requested the bailiffs who came to seize his only piece of furniture, (namely, the bed on which he lay), to spare him the use of it for a few hours, and then to remove him to the prison of the grave. Such was the melancholy end of a man of quiet unobtrusive merit, whose industry had been so great as to produce nearly two thousand four hundred prints, in every department of the art, portrait, history, costumes, antiquities, entomology, landscapes, &c. These are for the most part etchings, touched here and there with the graver. "His small plates are better than his larger, his style of handling his etching needle is pleasing, and was original; and in the accuracy, freedom, lightness, spirit, and finish, of his cities, abbeys, cathedrals, and some other of his landscapes, and his plates of natural history and still life, he much excelled his contemporaries, even those on the Continent; but in drawing the human figure he was not equally well informed, and his hands and feet are therefore often defective."

Francis Barlow, a native of Lincolnshire, was a very meritorious engraver of this period. His great merit lay in designing and etching quadrupeds, birds, fishes, and insects; and he became noted by his admirable embellishment of Æsop's fables. Several considerable works published about 1652 are illustrated by the engravings of this artist.

Section 7.

Prince Rupert's discovery of Mezzo-tinto.

The discovery of *mezzo-tinto* took place in the time of Barlow, and is thus noticed by Walpole: "It is a trite observation that gunpowder was discovered by a monk, and printing by a soldier. It is an additional honour to the latter profession to have invented mezzo-tinto. Few royal names appear at the head of discoveries; nor is it surprising. Though accident is the common mother of invention, yet genius being necessary to aid the casual production, and usher it into existence, one cannot expect that many of the least common rank should be blest with uncommon talents. Quickness to perceive, and sagacity to apply, are requisite to fortuitous discoveries. Gunpowder or printing might have fallen in many a prince's way, and the world still have been happy or unhappy enough not to possess those arts. Born with the taste of an uncle whom his sword was not fortunate in defending, Prince Rupert was fond of those sciences which soften and adorn a hero's private hours, and knew how to mix them with his minutes of amusement, without dedicating his life to their pursuit (like us, who, wanting capacity for momentous views, make serious study of what is only the transitory occupation of a genius). Had the court of the first Charles been peaceful, how agreeably had the prince's congenial propensities flattered and confirmed the inclinations of his uncle! How the muse of arts would have repaid the patronage of the monarch, when for his first artist she would have presented him with his nephew! How different a figure did the same prince make in a reign of dissimilar complexion! The philosophic warrior who could relax himself into the ornament of a refined court, was thought a savage mechanic when courtiers were only voluptuous wits. If the prince was defi-

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cient in the transient varnish of a court, he, at least, was adorned by the arts that polish, which alone can make a court attract the attention of subsequent ages." Of the discovery itself we gather the following account. The prince, on going out one morning at an early hour, observed the sentinel at some distance from his post, very busy doing something to his piece. The prince asked what he was about? He replied, the dew had fallen in the night, had made his fusil rusty, and that he was scraping and cleaning it. The prince on looking at it was struck with something like a figure eaten into the barrel, with innumerable little holes close together, like friezed work on gold or silver, part of which the fellow had scraped away. From this trifling circumstance the prince conceived the idea of mezzo-tinto. He concluded that some contrivance might be found to cover a brass plate with such a grained ground of fine pressed holes, which would undoubtedly give an impression all black, and that by scraping away proper parts the smooth superficies would leave the rest of the paper white. Communicating his idea to Wallerant Vaillant, a painter whom he maintained, they made several experiments, and at last invented a steel roller, cut with tools, to make teeth like a file or rasp, with projecting points which effectually produced the black grounds; those being scraped away and diminished at pleasure, left the gradations of light. This invention, by its softness and novel effect, produced great surprise and admiration among artists and others. After stating the merits of this discovery, Walpole remarks, "Curious as the discovery was, it did not produce all it seemed to promise; it has diversified prints, rather than improved them; and though Smith, who carried the art to the greatest height yet known, had considerable merit, mezzo-tintos still fall short of fine engravings. But before the secret passed into his hands, it was improved by Blooteling, who found out the application of the chisel for laying grounds, which much exceeded the roller. George White afterwards made use of the graver for forming the black spot in the eyes, and sharpening the light, which in preceding mezzo-tinto he observed had never been sufficiently distinct."

It was necessary to notice this discovery because of its close connexion with engraving, but we now resume our historical notice of the latter art, as it progressively advanced in England. William Faithorne is noted for the softness and force, freedom and finish, of his style, thus showing that the effect of chiaroscuro did not depend upon unblended masses of black and white. He lived in Hollar's time, and seems to have occupied the same house, and to have shared in many of the vicissitudes of that artist. While suffering imprisonment for his attachment to the royal cause, he still exerted his skill as an engraver; and produced, in Aldersgate Prison, among other works, a small head of the first Villiers, duke of Buckingham, in the circular style of Mellan. After much solicitation on the part of his friends, he was permitted to retire to France, where he found protection and encouragement. He returned to England about 1650, and opened a shop near Temple Bar, where he sold Italian, French, and Dutch prints, as well as the engravings of English artists. Here also he followed his profession for nearly thirty years, and produced many works of merit. Towards the close of his life he retired to a more private situation, in Printing House Yard, Blackfriars, where he died in 1691, and was buried in the church of St. Anne. The more important of this artist's works were his portraits. He worked almost entirely with the graver, in a free clear style, paying more attention to the beauty of his lines than most of his British predecessors. In the early part of his life he seems to have followed the Dutch and Flemish manner of engraving; but on his return from France, his improvement was evidently considerable. His drawing of the human figure is neither correct, nor in good taste, so that his historical works are not regarded as fair specimens of his talents. Portraiture was his favourite branch of study, and that to which he chiefly confined his attention. His son, William Faithorne, executed many portraits in mezzo-tinto; but they are very inferior to the works of his father.

Robert and George White are favourably noticed as engravers, and as practising with success the new art of mezzo-tinto.

The want of taste, and the dissolute manners of the court of Charles the Second, had the worst effect on the art of engraving. It would be useless to record the names of engravers whose works are, in many cases, of very low standing. Ingenuity was indeed conspicuous in some of the works of the period we allude to, but it was exercised in such senseless labour as that of executing, in the lines of the human face, a quantity of exceedingly fine writing not to be read without a magnifying glass, but, by that means, discovered to consist of the Lord's Prayer, ten Commandments, &c. But as that period, called the Augustan age, began to dawn upon England, the follies and vagaries of this art in preceding reigns, were nearly put to flight.

Section 8.

First Engravings from Raphael's Cartoons.

Dorigny, a Frenchman by birth, who had spent ten years in endeavouring to emulate the vigour and grandeur of Audran's style of engraving, came to England in 1711, with the design of engraving the Cartoons of Raphael. After many difficulties, he had apartments assigned to him, by the Queen, in Hampton Court Palace, and he commenced his work by public subscription, at four guineas the set. The labour of seven plates, of the large dimensions resolved on, seemed too much for one individual, and accordingly we find Dorigny seeking assistance from France. Dupuis and Dubosc were the names of the artists who came from Paris to lend him the necessary assistance, but both these artists left him before the expiration of the first three years.

On the 1st of April, 1719, after Dorigny had been employed for seven years in his work, he had the gratification of presenting two complete sets of these celebrated engravings (with a dedication) to George the First. He also gave one set to the Prince and another to the Princess. The present was acknowledged by a gift from the King of a purse containing one hundred guineas, and from the Prince of a medal; while the Duke of Devonshire, from whom the artist had borrowed four hundred pounds, freely remitted the interest for four years. These engravings extended the reputation of Dorigny throughout Europe, and in the succeeding year he received the honour of knighthood from the hands of the King.

Dorigny had begun to study drawing and engraving rather late in life, and he must have been about fifty years of age when he commenced the engravings from the cartoons. It is therefore the more surprising that so considerable a degree of success should have attended his labours. In this light it has been remarked that the power he acquired in the use of the graver was extraordinary. "The art with which he mingled the lines of the graver with those of the etching-needle is also much to be admired; and though it must be allowed that in copying Raffaelle's forms he has often lost much of their exquisite grace and chasteness, and has rendered the character of his heads but coarsely, yet on the whole he rides his graphic Pegasus with masculine grace. There is a manly energy and freedom in his style, bridled by simplicity; his shadows are full-toned, clear, and rich, and though his flesh be deficient in characteristic texture, the lines are often conducted over his draperies with unprecedented freedom and elegance, of which the figure of 'St. Paul preaching at Athens,' and that of the same Apostle in the cartoon of 'Elymas the Sorcerer struck blind,' may be sufficient examples."

While Dorigny was making his drawings of the cartoons a person in London offered him two hundred pounds for them; but the artist refused to make any agreement until all the plates should be completed. These drawings were afterwards sold at his auction for fifty-two guineas, and the total amount of his drawings realized three hundred and twenty pounds.

Passing by some intervening names, we arrive at that of George Vertue, whose influence on the art of engraving is best appreciated by considering the state into which that art had fallen previous to his day. Walpole describes the state of art in England as being then at the lowest ebb. The best performers were worn out; the war with France shut the door against recruits; national acrimony, and the animosity of faction, diverted public attention from common arts of amusement. Such was the state of things when our young engraver was recommended to the notice of Sir Godfrey Kneller, whose riches and influence enabled him to deal extensive patronage to any one who might be considered worthy of his favour.

George Vertue was born in the parish of St. Martin in the Fields, and was apprenticed at the age of thirteen to a man who engraved arms on plate, and had the chief business in London, but this man being obliged to leave the kingdom after Vertue had served him three or four years, the youth

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returned to his parents and practised drawing. He afterwards entered into an engagement with Michael Vandergutch for three years more, a term which he finally protracted to seven years, engraving copper-plates for him. In the year 1709 he began to work on his own account, and was employed for the first twelvemonth in drawing and engraving for books. The important protection of Sir Godfrey Kneller was at this period extended to him, and caused no small joy to the artist and his family. His father dying soon afterwards, a widow and several children were left to be supported by his industry alone. "I was the eldest," says Vertue, "and the only one that could help them, which added circumspection in my affairs then, as well as industry to the end of my life."

The merit of the artist soon extended his fame, and gave him the advantage of illustrious patronage. Lord Somers employed him to engrave a plate of Archbishop Tillotson, and this seems to have been the groundwork of his reputation. "Nothing like it," says Walpole, "had appeared for some years, nor at the hour of its production had he any competitors."

The Academy of Painting was instituted in 1711, and Vertue was one of the first members. Soon after the accession of George the First he published a portrait of the king, from a picture by Kneller, which being the first engraved likeness of that monarch many thousands were sold, though the merit of the performance was not great. In 1713 he began his researches and collections towards the great work which he afterwards published, entitled the *History of the Arts in England*. In order to perfect this work he made numerous journeys to different parts of England, generally in company with one or other of his generous patrons. In 1717 the Society of Antiquaries appointed Vertue their engraver. The University of Oxford also employed him for many years to engrave their almanacs, and these he greatly improved by introducing views of public buildings, and notices of historical events. In 1730 Vertue published his twelve heads of distinguished poets, an admirable work which he intended to have followed by the portraits of other eminent men arranged in classes. But the scheme was taken out of his hands, and the benefit lost to him. Vertue is supposed in some instances to have suffered on account of his rigid attention to truth. Where he could not discover he never supplied; but less scrupulous artists, when unable to obtain a genuine likeness of a great man of some remote period, would invent a portrait to supply the deficiency.

Vertue was employed three years in illustrating the translation of Rapin's *History of England*, which came out in folio numbers, and had a great sale.

The Earl of Oxford was a liberal and constant friend to our artist, and enabled him largely to indulge his taste for topographical antiquities, by taking him to many of the most interesting and remarkable spots in the kingdom. In 1741 he lost this noble friend and patron, and for two years he remained so much affected by his loss as to pursue but languidly his former avocations. "Death," he says, "put an end to that life which had been the support, cherisher, and comfort of many, many others, who are left to lament—but none more heartily than Vertue!"

The modesty and merit of this artist still raised him friends. The Duke of Norfolk gave him employment, and he was honoured by the notice of the Prince of Wales, who employed him in collecting prints and making catalogues, and who bought many of his own miniatures and prints. But here, as in the former case, death deprived him of his patron, and from the dejection occasioned by this melancholy event, the artist never wholly recovered. Vertue died in 1756, and was buried in the cloisters of Westminster Abbey. The talent of this artist has been doubtless exaggerated; but when we consider the state of art in his day, and the industry with which he exerted himself in its revival, we cannot think the praise of Walpole and others, either unmerited, or highly overwrought.

Section 9.

The Works of Hogarth.

We next arrive at Hogarth, of whom England is justly proud, both as a painter and engraver, and whose genius was of so bold and original a nature that all preceding names are at once cast in the shade. William Hogarth was born in London in 1697. He was driven from his father's profession, which was that of literature, by observing the difficulties and inconveniences which attended it. It was therefore quite in accordance with his own wishes, that he was taken from school, and placed on a long apprenticeship to a silver-plate engraver. But he says, "I soon found this business in every respect too limited. The paintings of St. Paul's Cathedral and Greenwich Hospital, which were at that time going on, ran in my head, and I determined that silver-plate engraving should be followed no longer than necessity obliged me to it. Engraving on copper was at twenty years of age my utmost ambition. To attain this it was necessary that I should learn to draw objects something like nature, instead of the monsters of heraldry; and the common methods of study being much too tedious for one who loved his pleasure, I was led to consider whether a shorter road than that usually travelled was not to be found. I had learned by practice to copy with tolerable exactness in the usual way; but it occurred to me that there were many disadvantages attending this method of study, as having faulty originals, &c. More reasons not necessary to enumerate, struck me as strong objections to this practice, and led me to wish that I could find the shorter path, fix forms and characters in my mind, and, instead of copying the lines, try to read the language, and, if possible, find the grammar of the art, by bringing into one focus the various observations I had made, and then trying by my power on the canvass and copper, how far my plan enabled me to combine and apply them to practice."

The success of the artist in this novel mode of procedure is well known, and must not be too largely dwelt on here, as it equally belongs to the history of painting. As soon as Hogarth became master of his own time, he began diligently to practice engraving on copper, and found plenty of employment in frontispieces to books, prints to *Hudibras*, &c. But the treatment he received from the booksellers soon compelled him to publish on his own account; but here again he had to encounter a monopoly of printsellers, "equally mean and destructive to the ingenious." He complains thus. "The first plate I published, called 'The Taste of the Town,' had no sooner begun to take a run, than I found copies of it in the print-shops, vending at half-price, while the original prints were returned to me again, and I was thus obliged to sell the plate for whatever these pirates pleased to give me,—as there was no place of sale but at their shops. Owing to this and other circumstances, by engraving, until I was near thirty, I could do little more than maintain myself."

When Hogarth was thirty-two years of age, he married the daughter of Sir James Thornhill. For a few years after this event he employed himself in painting small conversation pieces, which succeeded well for a time, but were not permanently successful, nor sufficiently profitable to pay the expenses of his family. "I therefore," says he, "turned my thoughts to a still more novel mode, *i.e.*, painting and engraving moral subjects, a field not broken up in any country or age. I thought that both writers and artists had overlooked the intermediate species of subject, which may be placed between the sublime and the grotesque; I therefore wished to compose pictures on canvass similar to representations on the stage."

Having had his plates pirated in almost all sizes, Hogarth applied to parliament, in 1735, for redress. This he obtained in so liberal a manner as to remove further ground of complaint, and greatly to increase the sale of prints in this country. The copyright of prints being secured by this act, not only were the artists and the public benefited by it, but also the printsellers themselves, who were now secured in the possession of such engravings as they might purchase. Yet Hogarth says, "The dealers in prints and pictures found their craft in danger, by what they called a new-fangled innovation. Their trade of living and getting fortunes by the ingenuity of the industrious has, I know, suffered much by my interference; and if the detection of this public band of cheats and oppressors of the rising artists, be a crime, I confess myself most guilty."

The fame of our artist had now been spread far and wide by his celebrated series of pictures, pointed with such strong satire at the vices of the age. His "Harlot's Progress," in six plates, and his "Rake's Progress," in eight, excited a great sensation. At this time Swift wrote those oft-quoted lines—

> How I want thee, hum'rous Hogarth!
> Thou, I hear, a pleasant rogue art;
> Were but you and I acquainted
> Every monster should be painted:
> You should try your graving tools
> On this odious group of fools [the Legion Club],
> Draw the beasts as I describe 'em,
> Form their features, while I gibe 'em.

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The next productions of this artist were engravings of the "Sleeping Congregation," the "Distressed Poet," and the "Arms of the Undertakers' Company." In 1737 he produced "The Lecture," and in the following year, "Morning," "Noon," "Evening," and "Night." The "Salisbury" flying coach, in the engraving of "Night," is supposed to be a satire on a nobleman who delighted to drive his own horses. Other forcible and satirical productions followed. "The Enraged Musician," "Taste in High Life," and the six plates entitled "Marriage à la Mode," appeared before the year 1745. "The Idle and Industrious Apprentices," in twelve plates, designed and engraved by Hogarth, appeared in 1747. Of this series Hogarth himself writes, "The effects of idleness and industry exemplified in the conduct of two fellow-'prentices. These twelve prints were intended for the instruction of young people, and considering the persons they were intended to serve, I have endeavoured to render them intelligible and cheap as possible." (The twelve were originally published at the low price of twelve shillings.)

In 1749 Hogarth produced "The Gate of Calais," or "Roast Beef of Old England." In the same year he also engraved and published his own portrait in a cap, with his favourite pug-dog, and palette, on which is drawn that mysterious "line of beauty and grace" which has excited so much attention. "The March to Finchley" appeared in 1750. By this work Hogarth meant no more than a humorous piece of satire, though it is said to have given offence in some quarters. On the approach of Prince Charles in 1746 the Guards were ordered to march to Finchley, in order to extinguish the Scottish rebellion, but Hogarth has given a ludicrous air to their departure from London. The scene is laid in Tottenham Court Road. In the distance the more orderly and obedient portion of the soldiery are seen marching northward; but in the rear is a baggage-waggon, lumbering along with its burthen of women, babies, knapsacks, and camp-kettles, and around it is a disorderly torrent of soldiers, inflamed with liquor. "There is such staggering and swaggering, such carousing and caressing, such neglect of discipline, and obedience to nothing save the caprice of the moment, and yet all is natural, consistent and characteristic."

The original picture was presented by the artist to the Foundling Hospital, or, as Cunningham asserts, was disposed of by lottery. Two pictures, called "Gin Lane," and "Beer Street," appeared the following year, and of these we have Hogarth's own account as follows:—"When these two prints were designed and engraved, the dreadful consequences of gin-drinking appeared in every street. In 'Gin Lane,' every circumstance of its horrid effects is brought to view *in terrorem*. Idleness, poverty, misery, and distress, which drive even to madness and death, are the only objects that are to be seen; and not a house in tolerable condition but the pawn-broker's and gin-shop. 'Beer Street,' its companion, was given as a contrast, where that invigorating liquor was recommended in order to drive the other out of vogue. Here all is joyous and thriving, &c." Hogarth next published his "Four Stages of Cruelty" and his remarks on them do honour to his humane and generous feelings. "These prints," he says, "were engraved with the hope of, in some degree, correcting the barbarous treatment of animals, the very sight of which renders the streets of our metropolis so distressing to every feeling mind. If they have had this effect, and checked the progress of cruelty, I am more proud of having been the author, than I should be of having painted Raphael's cartoons." The execution of this well-intended work was however such that we find Cunningham, in his *Life of Hogarth*, saying, "I wish it had never been painted. There is indeed great skill in the grouping, and profound knowledge of character; but the whole effect is gross, brutal, and revolting. A savage boy grows into a savage man, and concludes a career of cruelty and outrage by an atrocious murder, for which he is hanged and dissected. The commencement is painful: and the conclusion can scarcely be looked upon except by men practised in surgery, or the shambles."

Our artist now published his famed *Analysis of Beauty*, written with the view of fixing the fluctuating ideas of taste, and embellished with two curious illustrative engravings. In 1754 appeared the first of four plates containing an admirable satire on "Parliamentary Elections." These plates were entitled, "An Election Entertainment," "Canvassing for Votes," "The Polling," and "Chairing the Members." In 1758 appeared "The Bench," and in the following year "The Cockpit." Of the latter of these Ireland says:—"The scene is probably laid at Newmarket, and in this motley group of peers, pickpockets, butchers, ratcatchers, gentlemen, and gamblers, Lord Albemarle Bertie, being the principal figure, is entitled to precedence. What rendered his lordship's passion for amusements of this description very singular, was his being totally blind. In this place he is beset by seven steady friends, five of whom offer to bet with him at the same instant on the event of the battle. One of them, a lineal descendant of Filch, taking advantage of his blindness and negligence, endeavours to convey away a bank-note deposited in our dignified gambler's hat, into his own pocket; of this attempt his lordship is apprised by a ragged pot-boy, and an honest butcher, but he is so much engaged in uttering these important words 'Done! done! done!' that he cannot attend to their hints, and it seems more than probable that the stock will be transferred, and the note negotiable, in a few seconds."

A few other satirical works, such as the *Times*, *Bathos*, &c., closed his career, and in October, 1764, he died at the age of sixty-seven, and was buried at Chiswick. Walpole observes of this wonderful man: "It is to Hogarth's honour that in so many scenes of satire or ridicule, it is obvious that ill-nature did not guide his pencil. His end is always reformation, and his reproofs general. Except in the print of the 'Times,' and two portraits of Wilkes and Churchill, no man amidst such a profusion of characteristic faces, ever pretended to discover or charge him with the caricature of a real person; except of such notorious characters as Chartres, and mother Needham, and a very few more, who are acting officially and suitably to their professions. As he must have observed so carefully the operation of the passions on the countenance, it is even wonderful that he never, though without intention, delivered the very features of any identical person. It is at the same time a proof of his intimate intuition into nature: but had he been too severe, the humanity of endeavouring to root out cruelty to animals would atone for many satires. It is another proof that he drew all his stores from nature, and the force of his own genius, and was indebted neither to models nor books for his style, thoughts, or hints, that he never succeeded when he designed for the works of other men. He could not bend his talents to think after anybody else. He could think like a great genius, rather than after one."

Section 10.

The Contemporaries of Hogarth.

At the same period with Hogarth flourished several meritorious engravers, some of whom assisted him in his numerous works. Arthur Pond and Simon Francis Ravenet were among the number. The latter instructed both Ryland and Hall in the art of engraving. Several works were executed with great spirit and taste, by Arthur Pond and George Knapton, particularly a set of plates, in imitation of chalk, and washed drawings, from the designs of the great Italian masters: some of these are in chiaroscuro with etched outlines. These artists flourished about 1740.

William Wynne Ryland was born in London in 1732, and showed an early genius for the fine arts. He studied first under Ravenet, and then at Paris under Boucher, who has been called the misleader of the taste of France, and to whom we must perhaps attribute the perversion of Ryland's talents. He acquired a habit of "stippling" with the graver, by which process he gave the same effect to his engravings which the custom of "stumping" in pencil drawings is well known to produce. Thus all energy of style was destroyed, and an injury done to the art. He was appointed engraver to the King, and his new manner of engraving was prized merely for its novelty, the public eagerly seeking after his prints, without waiting to consider whether the change of style made them better representations of nature. Ryland suffered for forgery, August 29, 1783. John Hall was an artist of some eminence. His "William Penn treating with the Indians," and "Cromwell dissolving the Long Parliament," both after West, are good specimens of his bold clear style.

Chatelain was a British artist, whose abilities were of a first-rate order, but he was so idle and dissolute that he seldom exerted himself, except when pressed by necessity to do so. He engraved after Gaspar Poussin and others, for a collection of fine landscapes, published by Boydell in 1744.

Boydell was a spirited publisher, as well as an engraver, He was brought up to the business of a land surveyor, but was so attracted by a book of prints, by Toms, an engraver of architectural subjects, that he proceeded to London at the age of twenty-one, and bound himself as

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a pupil to that artist for six years. At the expiration of that time, he engraved and published several views of his own, in one volume for five guineas. From this beginning he rose to considerable wealth, and he was wont to remark, in after times, that his was "the first book that ever made a Lord Mayor of London." He managed his extensive commerce in prints throughout Europe with so much judgment and liberality, that he had the opportunity of employing the ablest artists of his time, and of exporting largely, and in an advantageous manner, that which had formerly been an article of import. Boydell's property was greatly injured by the French Revolution, so that his celebrated work, *The Shakspeare Gallery*, which he had intended to bequeath to the public, was sold. He served the office of Lord Mayor in 1791, and died in 1804.

Vivares is celebrated as being the first artist of any school who was completely successful in rendering with fidelity and spirit the landscapes of the great Italian masters. His best works are from Claude. He learned his art rather late in life, from Chatelain, but improved on the style of his master, and became remarkable for the delicacy and lightness of his foliage, and for his broad, clear effect. He acquired much greater freedom in etching than had been before possessed, and bringing his plates to a state of considerable finish with the point, he put on the last touches with the graver.

The same mode of proceeding was pursued by Woollett, whose landscapes are considered to present perfect models of style. He has chiefly engraved from Wilson and others; but he did not neglect historical subjects, as we find by his "Death of General Wolfe" after West, and some others. "In tracing the progress of this branch of art," says a writer in the *Edinburgh Encyclopædia*, "we cannot withhold our admiration from these two great men, who, from the state of total insignificance and neglect in which they found it, raised it at once to such dignity and perfection. The earliest landscapes we have of any importance from the works of the great painters, are the engravings of Bolswert, after the pictures of Rubens. They are executed with his usual ability, but being entirely with the graver, give a very inadequate idea of the character of the foliage, the quality of surface of the several objects, or the brilliancy and splendour of Rubens. The large set of landscapes by Audran, after Nicolo Poussin, exhibits a nearer approximation to the true style of landscape, being executed with much greater delicacy, but from the unmanageable nature of the graver with which they are entirely executed, they are very defective in character and spirit. The discrimination of Vivares and Woollett pointed out the defects of their predecessors; and more particularly in the works of the latter, we find all the truth of nature united to all the beauties of mechanical skill."

John Browne was among the best landscape engravers of this period. Several large works after Salvator Rosa, Both, and other great masters, were executed by him in excellent style. He likewise etched many plates, which were afterwards finished by Woollett with the graver.

Artists in every department now rapidly multiplied in England. Among these Sir Robert Strange holds a high rank among engravers, and deserves especial notice. This eminent artist was born 1721 at Pomona, one of the Orkney Islands. After receiving a good education, he was invited to the study of the law; that profession not suiting his talents, and while yet uncertain what profession to choose, he went on board a man-of-war; but he soon became disgusted with a sea-life and returned home. In the mean time his brother had accidentally discovered in his drawer a variety of drawings and unfinished sketches, with which he appears to have amused those hours that his friends supposed were devoted to severer labours. These first essays of genius being shown to Mr. Richard Cooper, a gentleman who practised the fine arts at Edinburgh, they were highly approved, and he at once proposed that young Strange should be placed under his tuition. This was accordingly done, and the pupil was making rapid progress, when his studies were interrupted by the arrival of the young Chevalier in Scotland. Strange, being urged by the desire to gain the hand of Miss Isabella Lumisden, at her request joined the rebel army, and continued to serve until the total defeat of the Pretender's troops at Culloden. He lay concealed in the Highlands during some months, and when the vigilance of pursuit was somewhat abated, he visited Edinburgh, and began to earn money by the sale of portraits of the rival leaders in the rebellion, which he disposed of privately at a guinea each. Having married the lady above referred to, Strange proceeded to France to complete his studies. At Rouen he gained the prize for design. At Paris he studied engraving under the direction of the celebrated Le Bas. From this master he had the first hint of the use of dry point, which he afterwards greatly improved, and thus added superior beauties to his engravings.

In 1751 he settled in London, and devoted himself to historical engraving. He soon established a high reputation in this difficult style of his art, of which he is considered the father. In 1760 he visited Italy, where he was known through the medium of his engravings, and was received with honour and respect. He is the only Englishman whose portrait is introduced in the painting in the Vatican of the "Progress of Engraving."

The authority last quoted says of this artist, "He is admirable for the breadth of his effect, and the beauty of his execution; but his great excellency is the beauty and softness of his female flesh. In this last, notwithstanding the perfection that modern art has arrived at, in all those great qualities that result from mechanical skill, he has never been equalled by any master, as his engravings from the works of Titian, Guido, Correggio, and the other painters of the Italian schools, sufficiently show; but it is deeply to be regretted that, with so many excellencies, his drawing should be so incorrect, particularly in the extremities." He was appointed engraver to George the Third, and received the honour of knighthood in 1787. He died on the 5th July, 1792.

In landscape, besides Vivares, Woollett, and Browne, many fine works from the old masters, by Byrne, Mason, Wood, Elliott, Lowry, Wilson, Major, Earlom, and others, show the high standing of the English school; but in our own times the extent to which works on almost every subject are illustrated, is perhaps no real token for good to the art. Artists are indeed multiplied, and there is an unexampled demand for the fruits of their industry; but the study of the old masters is comparatively neglected, while topography, antiquities, &c., usurp so large a share of study and attention.

Among those who have illustrated the beautiful scenery of this country, and shown that it is quite as capable of pictorial effect as that of any foreign clime, we find one of the first to be Paul Sandby, born at Nottingham, in 1732. His early skill is evinced by the fact, that at the age of sixteen he was employed as draughtsman under Mr. David Watson, to complete a survey of the north and west portions of the Highlands, for the late Duke of Cumberland. While thus occupied among romantic scenery, he employed the leisure he could get from drawing plans, in making sketches of the wild and picturesque spots he met with. These he afterwards etched in small plates. Soon afterwards he was introduced to the notice and patronage of Sir Joseph Banks, and Sir Watkins William Wynne. With the former he made a tour through North and South Wales, and by the latter he was employed to design the most striking features of Welsh scenery. These views were afterwards engraved in aquatinta. Sandby was one of the original Members of the Royal Academy.

Of the engravers of the present day we purposely abstain from giving an account. Perhaps their art was never more highly prized, or more generally appreciated and encouraged. May this induce artists to take a high aim, and to employ the improved facilities and advantages they enjoy, in endeavouring to elevate and refine the public taste. It is not often that we meet with men who unite the talents of the painter and the engraver; yet it is much to be wished that the number of such individuals should increase. The advantage to an artist of being able to engrave his own pictures is a most invaluable one. It is impossible that another mind should grasp every idea intended to be conveyed by the artist, and grievous must be the disappointment in many cases, when in the engraving the painter seeks in vain for the spirit which he intended to pervade his work. The possibility of uniting the two arts, is manifest in the case of Hogarth, and has also been proved in our own time. May its desirability become more apparent, and influence the practice of our men of talent!

END OF THE TWENTY-THIRD VOLUME.

LONDON: JOHN W. PARKER, PUBLISHER, WEST STRAND.

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